JeffMD
Lower-Tract GU Malignancies: Prostate, Bladder and Testis Cancers

Mihir S. Shah, MD (mihir.shah@jefferson.edu)

Department of Urology

Sidney Kimmel Medical College

Thomas Jefferson University

No Disclosures

Learning Objectives

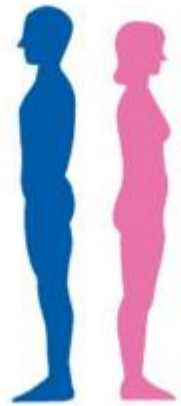
1. Explain the presented epidemiology, etiology of and risk factors for prostate cancer.
2. Discuss the role of physical exam and PSA in screening for prostate cancer.
3. Describe the process for diagnosis of prostate cancer and explain the diagnostic and prognostic characteristics of Gleason Score for prostate cancer.
4. Describe the treatment options for a patient who presents with localized prostate cancer.
5. Describe the treatment concepts for metastatic castrate sensitive and castrate resistant prostate cancer.
6. Explain the presented epidemiology of and risk factors for bladder cancer.
7. Describe the prognostic and therapeutic differences between muscle-invasive and non-muscle-invasive urothelial carcinoma of the bladder
8. Describe the epidemiology of and risk factors for testis cancer.
9. Describe the presentation, basic components of staging, treatment options, and outcomes for a patient who presents with testis cancer.
10. Identify the most common histologies and tumor markers of testis cancer

- Please note that for the purpose of this discussion the terms “male”, “men”, “women” and “female” all refer to biological sex and not gender identity.

2020 Cancer Statistics

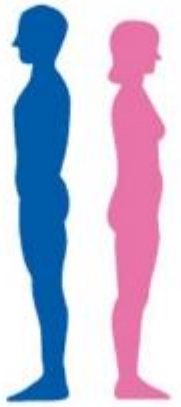
Estimated New Cases

Males			Females		
Prostate	191,930	21%	Breast	276,480	30%
Lung & bronchus	116,300	13%	Lung & bronchus	112,520	12%
Colon & rectum	78,300	9%	Colon & rectum	69,650	8%
Urinary bladder	62,100	7%	Uterine corpus	65,620	7%
Melanoma of the skin	60,190	7%	Thyroid	40,170	4%
Kidney & renal pelvis	45,520	5%	Melanoma of the skin	40,160	4%
Non-Hodgkin lymphoma	42,380	5%	Non-Hodgkin lymphoma	34,860	4%
Oral cavity & pharynx	38,380	4%	Kidney & renal pelvis	28,230	3%
Leukemia	35,470	4%	Pancreas	27,200	3%
Pancreas	30,400	3%	Leukemia	25,060	3%
All Sites	**893,660**	**100%**	**All Sites**	**912,930**	**100%**



Estimated Deaths

Males			Females		
Lung & bronchus	72,500	23%	Lung & bronchus	63,220	22%
Prostate	33,330	10%	Breast	42,170	15%
Colon & rectum	28,630	9%	Colon & rectum	24,570	9%
Pancreas	24,640	8%	Pancreas	22,410	8%
Liver & intrahepatic bile duct	20,020	6%	Ovary	13,940	5%
Leukemia	13,420	4%	Uterine corpus	12,590	4%
Esophagus	13,100	4%	Liver & intrahepatic bile duct	10,140	4%
Urinary bladder	13,050	4%	Leukemia	9,680	3%
Non-Hodgkin lymphoma	11,460	4%	Non-Hodgkin lymphoma	8,480	3%
Brain & other nervous system	10,190	3%	Brain & other nervous system	7,830	3%
All Sites	**321,160**	**100%**	**All Sites**	**285,360**	**100%**



Prostate Cancer

Prostate Cancer

- Epidemiology\Risk Factors
- Prostate Cancer Biology
- Diagnosis
- Prevention
- Treatment

Prostate Carcinoma

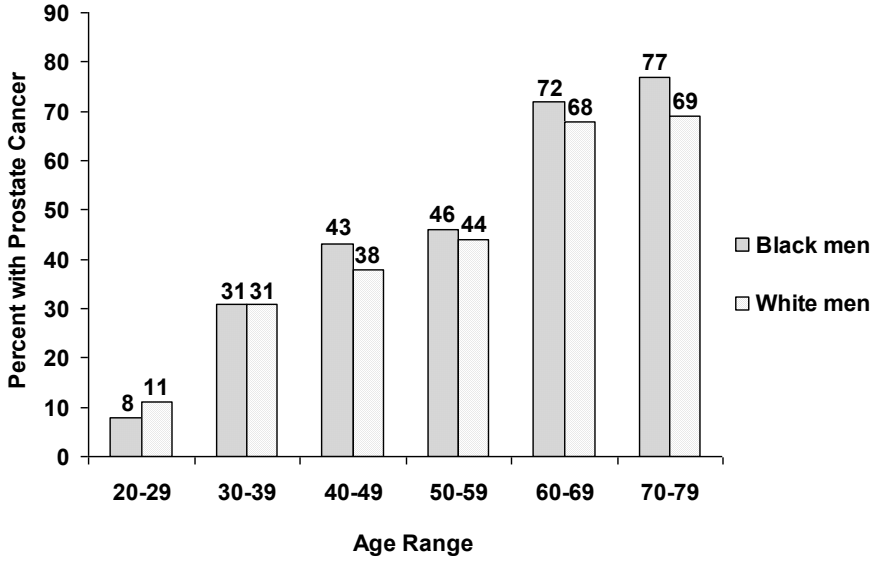


Prostate Cancer	Breast Cancer
• Most commonly diagnosed cancer in men	• Most commonly diagnosed cancer in women
• 1 in 6 men will be diagnosed	• 1 in 7 women will be diagnosed
• Over 230,000 new cases diagnosed annually	• Over 210,000 new cases diagnosed annually
• Accounts for 33% of all cancers	• Accounts for 32% of all cancers
• 5 year survival with early detection=100%	• 5 year survival with early detection=98%

"The two diseases have similar risk factors, overlapping treatment pathways and comparable implications for emotional health and intimate relationships."

Marc Shuman, MD, UCSF

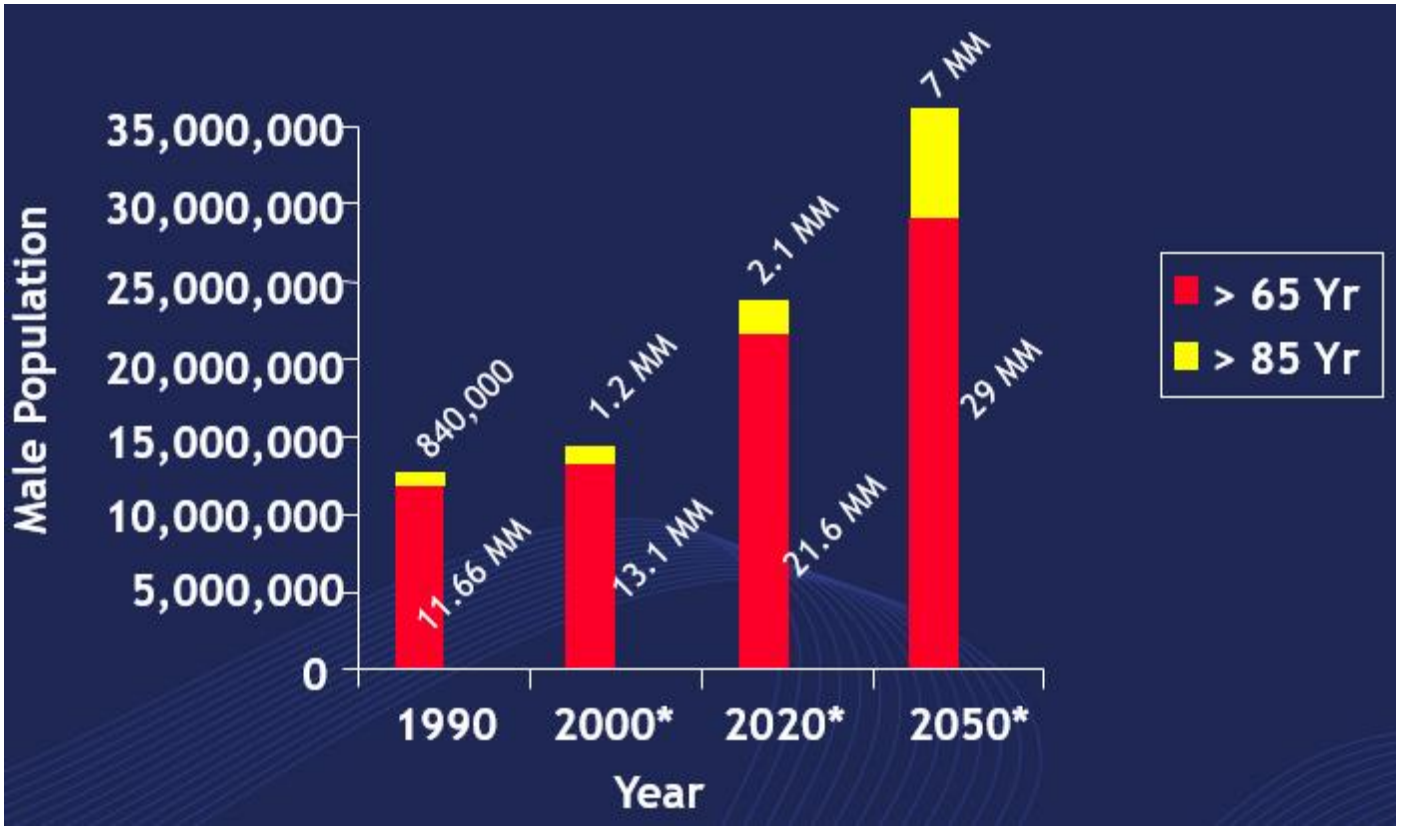
Risk of Prostate Cancer by Age and Race



Probability of Developing PC	
Birth to 39	1 in 9,4422
40 to 59	1 in 41
60 to 69	1 in 16
70 and older	1 in 8
Birth to death	1 in 7

Jemal et al., CA Cancer J Clin. 2010 Sep-Oct;60(5):277-300

Population Trend in the US Challenges Prostate Cancer



~80% of new prostate cancer diagnoses are in men >age 65yrs

Prostate Carcinoma
Etiology is multifactorial

- Androgens
- Heredity
- Environment

Androgens

- Role of hormones is essentially permissive
  - People who are castrated before puberty do not develop prostate cancer
  - Cancers respond (for a time) to chemical or surgical castration

Heredity

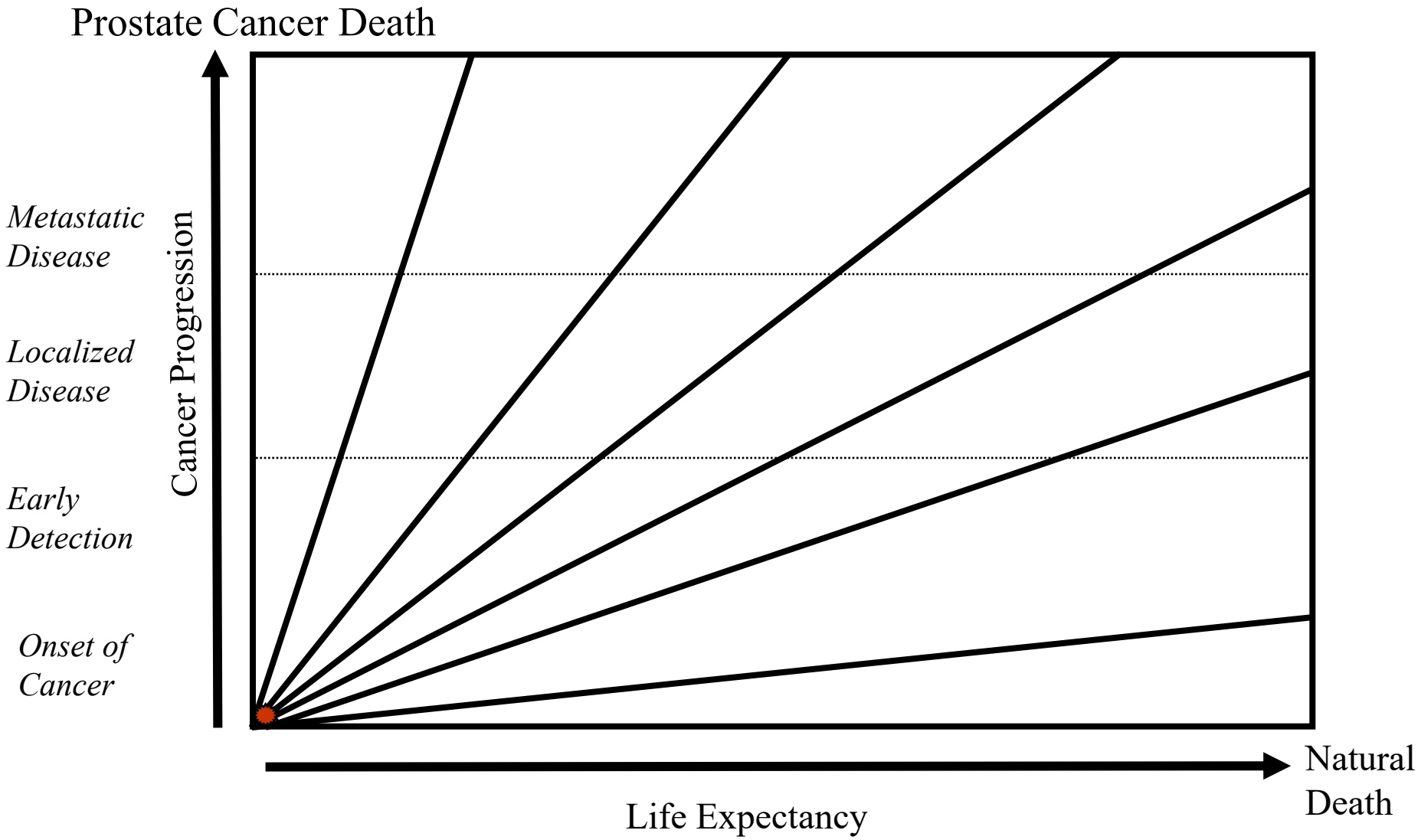
- Increased risk among first degree relative with prostate cancer
- Highest in Blacks/Scandinavian
- Lowest in Asians
- 2 or 3 first degree relatives increases the risk 5 to 11 fold
- Increased risk in relatives with breast cancer

Colloca et al. Acta Oncol. 2011 Jan;50(1):14-24.

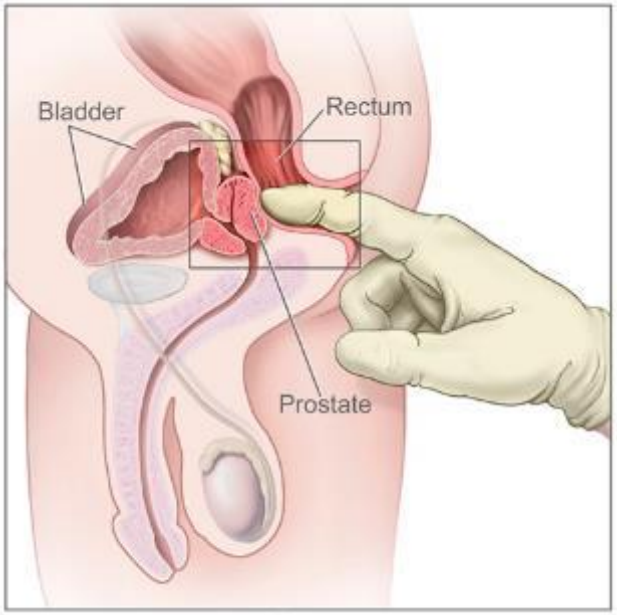
Environment

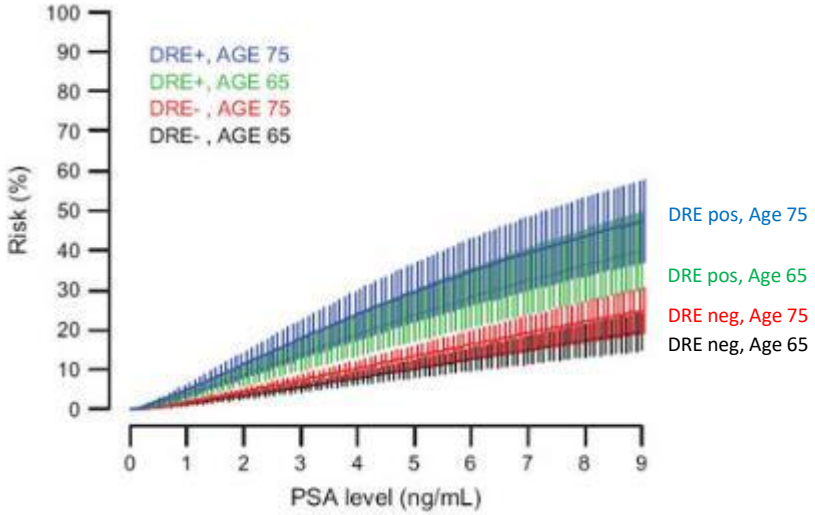
- Dietary
- Asian immigrants have higher risk (not as high as native born Americans)
- As Asian diet is “westernized” incidence appears to be rising.
- Specific dietary components have not been identified (fats?)

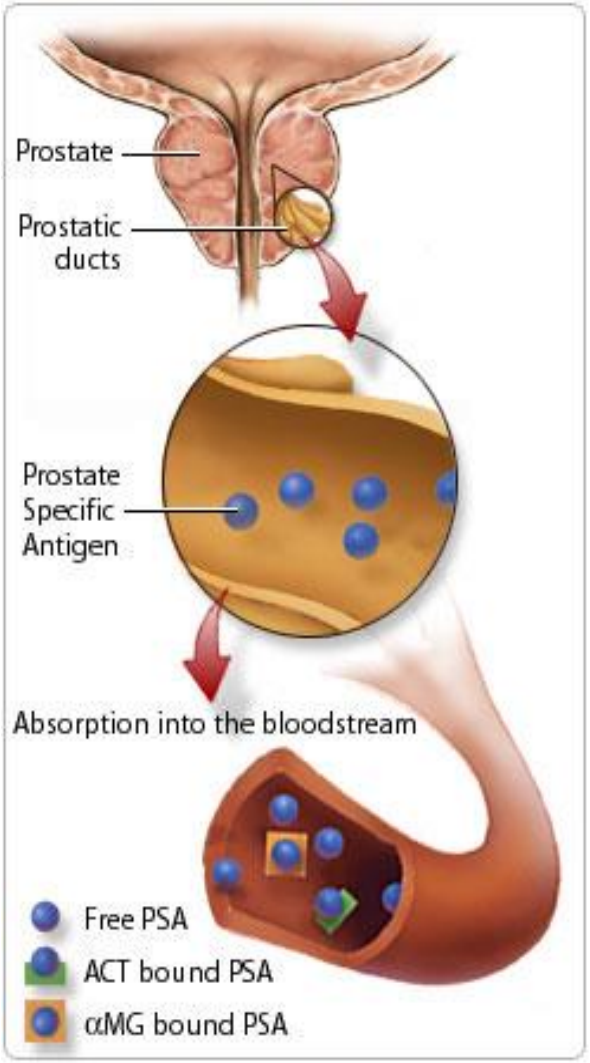
Prostate Cancer: Growth Rate and Progression



Diagnosis


Bladder
Rectum
Prostate


100
90
80
70
60
50
40
30
20
10
0
Risk (%)
DRE+, AGE 75
DRE+, AGE 65
DRE- , AGE 75
DRE- , AGE 65
0
1
2
3
4
5
6
7
8
9
PSA level (ng/mL)
DRE pos, Age 75
DRE pos, Age 65
DRE neg, Age 75
DRE neg, Age 65


Prostate
Prostatic ducts
Prostate Specific Antigen
Absorption into the bloodstream
Free PSA
ACT bound PSA
αMG bound PSA

Interpreting PSA

- PSA is a single chain serine protease
  - Member of the human tissue kallikrein family

ELEVATED:

Prostatitis, BPH, cancer, urethral/prostatic trauma, sex/ejaculation (rarely) (routine DRE does not)

DECREASED:

Antibiotics will lower PSA in 30% if infected

Simple repeat will lower in many men

Statins lower by ~ 4%

Finasteride/Dutasteride lower PSA by 50% (any dose)

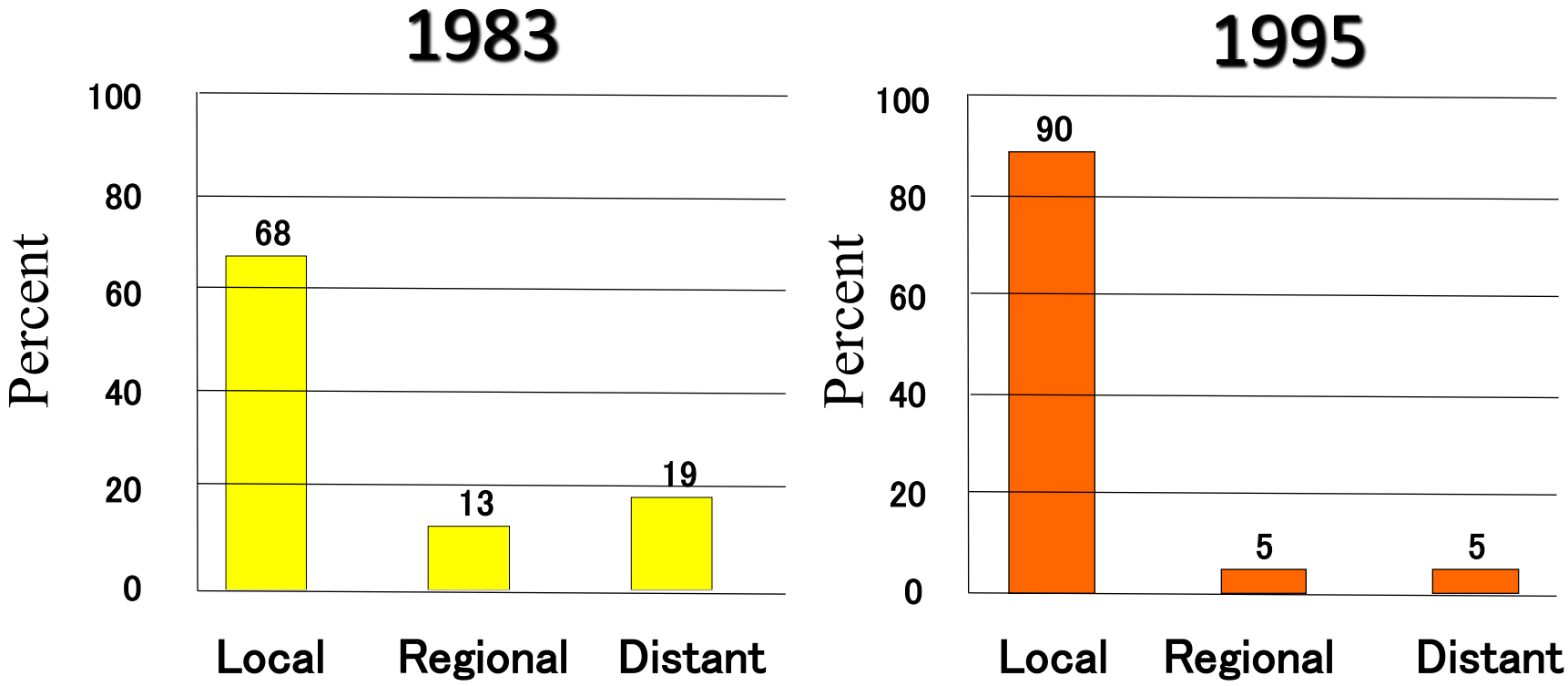
Assay variability can be as much as 20%

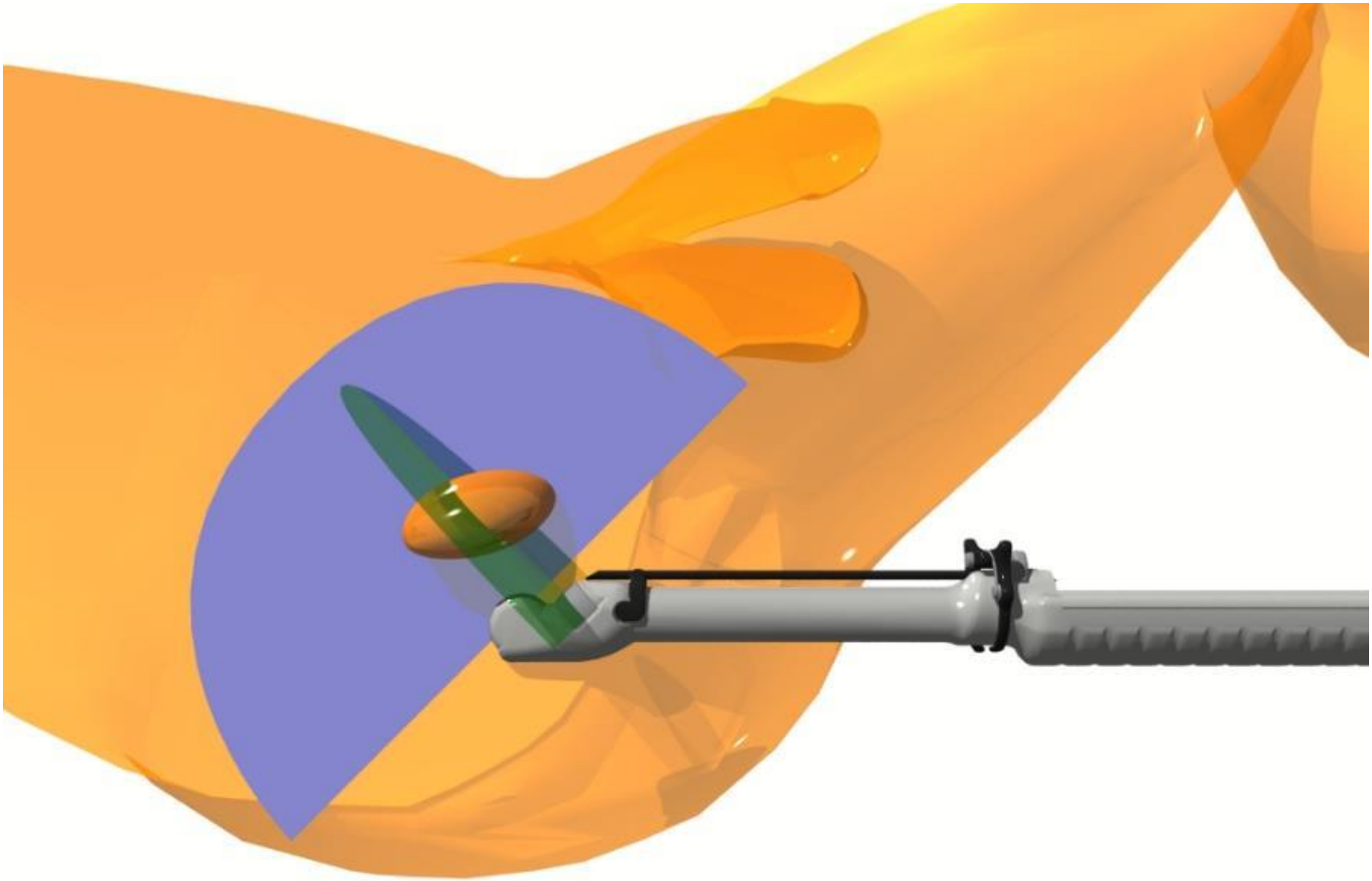
WHO standards 20-25% lower than Hybritech standard

No lower PSA limit eliminates cancer risk (PCPT trial)

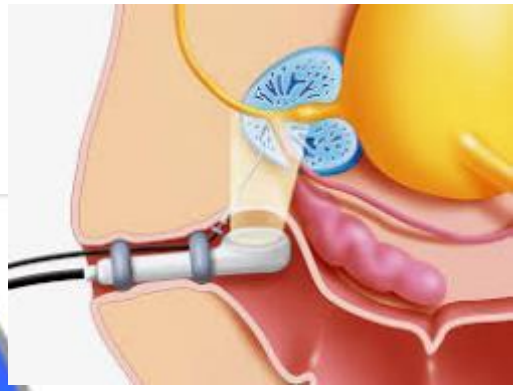
Shift in Stage at Diagnosis after Introduction of PSA into U.S.

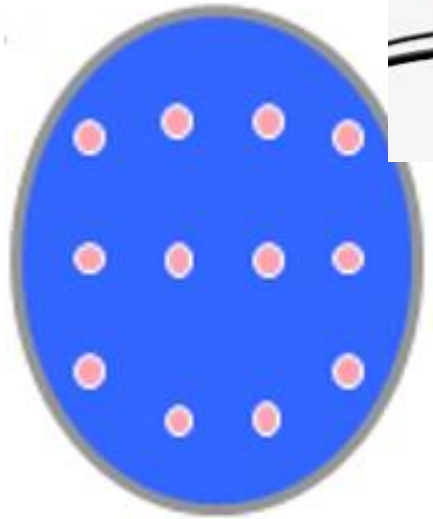
(SEER Data)



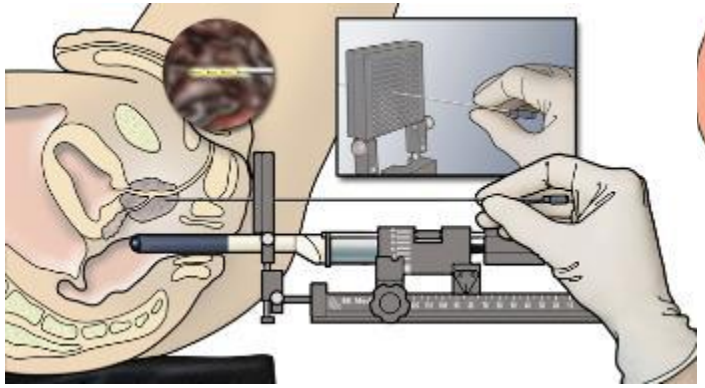


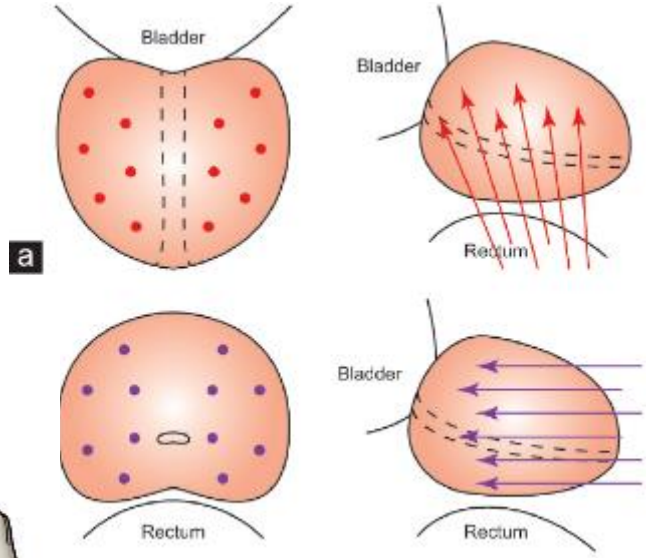
Ultrasound and Prostate Biopsy Transrectal vs Transperineal





Transrectal

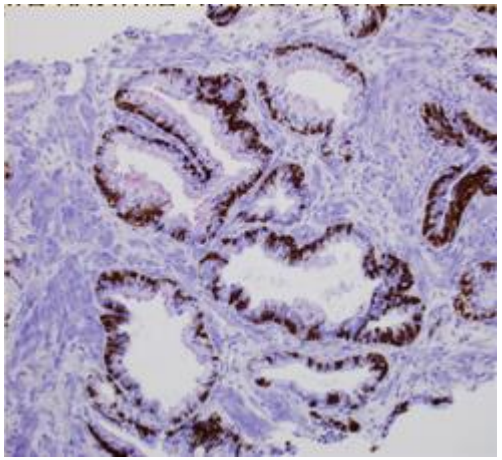




Transperineal

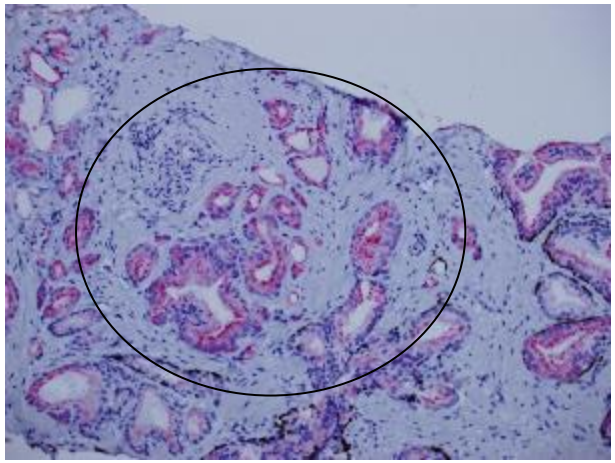
Immunohistochemical Stains for Cancer Tissue Diagnosis

BENIGN



- Normal glands have preserved basal cells (+BROWN)

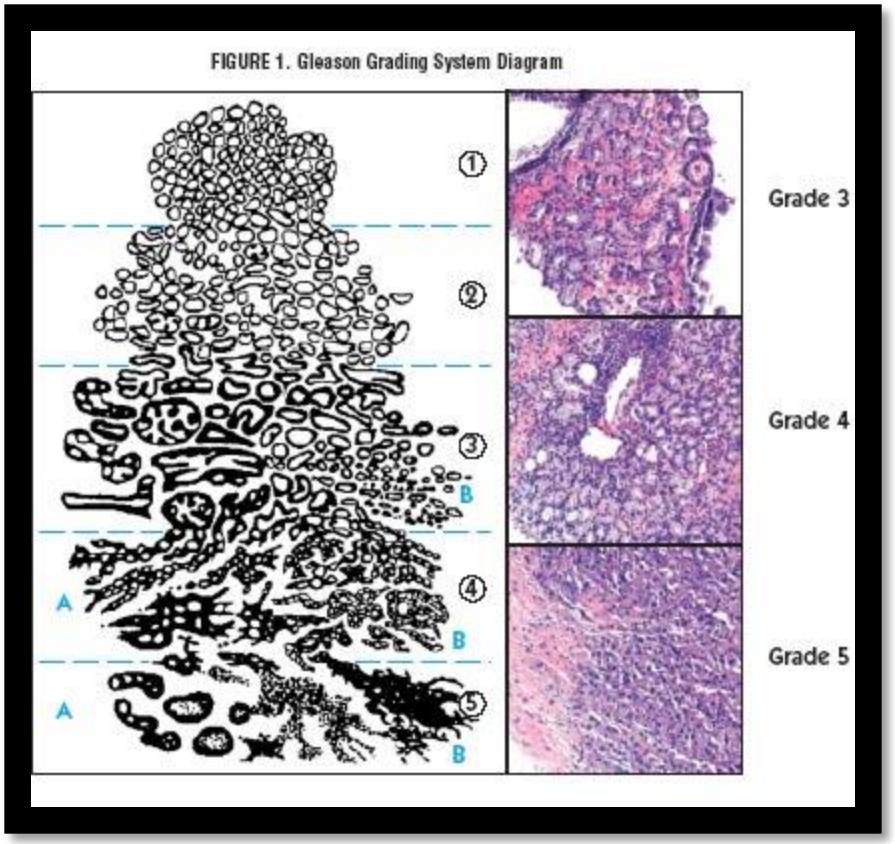
CANCER

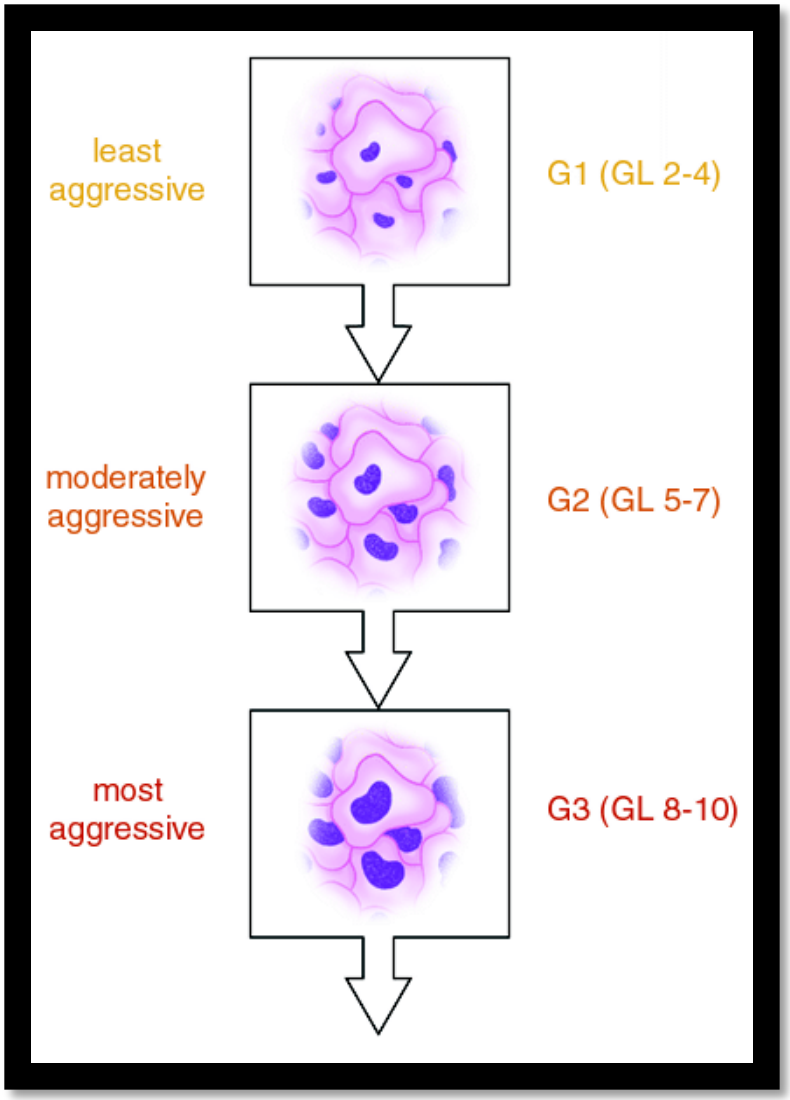


- Alpha-methylacyl-CoA racemase (AMCAR/p504s) is present in prostate cancer and HGPIN (+RED)

- Basal cell layer markers are absent in cancer and present in benign glands and HGPIN (-BROWN)

“Need to have Tissue for a Diagnosis” Gleason Score





Prognostic "Grade Group" Grading System

- **Grade Group 1** (Gleason score ≤6) – Only individual discrete well-formed glands histologically
- **Grade Group 2** (Gleason score 3+4=7) – Predominantly well-formed glands with a lesser component of poorly-formed/fused/cribriform glands
- **Grade Group 3** (Gleason score 4+3=7) – Predominantly poorly-formed/fused/cribriform glands with a lesser component of well-formed glands†
- **Grade Group 4** (Gleason score 8)
- **Grade Group 5** (Gleason scores 9-10) – Lacks gland formation (or with necrosis) with or w/o poorly-formed/fused/cribriform glands

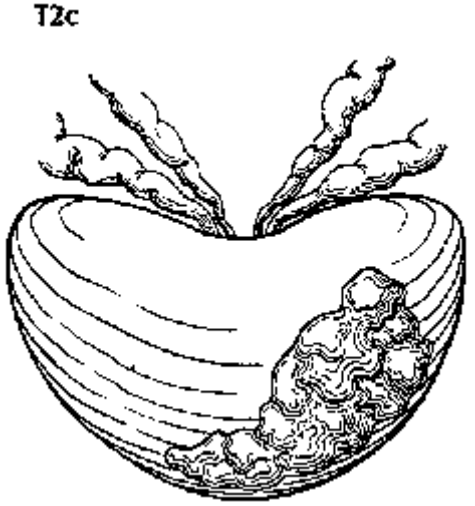
Clinical Staging of Prostate Cancer (TNM)

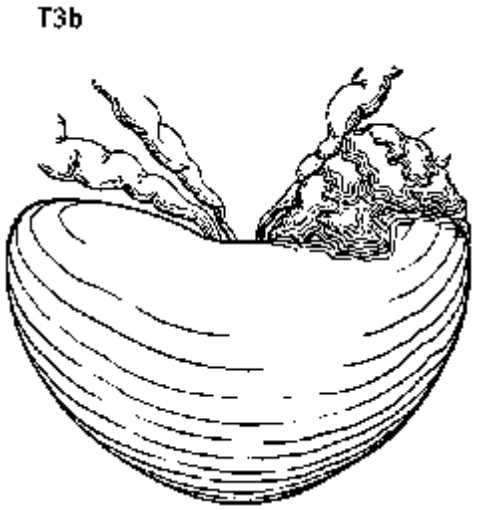
PSA and DRE

- Stage T1: not clinically detectable, only by core biopsy (T1c- elevated PSA alone)
- Stage T2: confined to the prostate
- Stage T3: with extracapsular penetration, prostatic bladder neck or seminal vesicle invasion
- Stage T4: invades adjacent structures like bladder, rectum, pelvic wall

CT and Bone Scan

- Stage N1: Lymph nodes metastasis
- Stage M1: Distant metastasis





Pathologic Staging of Prostate Cancer (TNM)

pT2: Organ confined

pT3: Extraprostatic extension
- pT3a: Periprostatic/ bladder neck invasion
- pT3b: Seminal vesicle invasion

pT4: Invasion of rectum, levator muscles and/or pelvic wall

pNX: Cannot be assessed

pN0: No regional lymph node metastasis

pN1: Metastasis in regional lymph node or nodes

pMx/M0

M1: Distant metastasis
- pM1a: Nonregional lymph node(s)
- pM1b: Bone(s)
- pM1c: Other site(s) with or without bone disease

Risk Stratification and Treatment
Active Surveillance

- **Very low risk**: T1c, not N1 or M1; Gl ≤ 6; fewer than 3 cores positive with cancer; ≤ 50% cancer in any 1 core
- **Low risk**: T1c, not N1 or M1; Gl ≤ 6; and PSA <10 ng/ml; but not meeting the very low risk criteria
- **Intermediate risk**: T1-2, not N1 or M1; Gl ≤ 7; and/or PSA 20 ng/ml; and not meeting very low or low risk criteria
- **High risk:** T1-4, not N1 or M1; Gl ≥ 8; and/or PSA <100 ng/ml; not meeting other risk groups criteria

Godtman et al. European Urol 63, 2013

Robotic-assisted Prostatectomy(RALP)





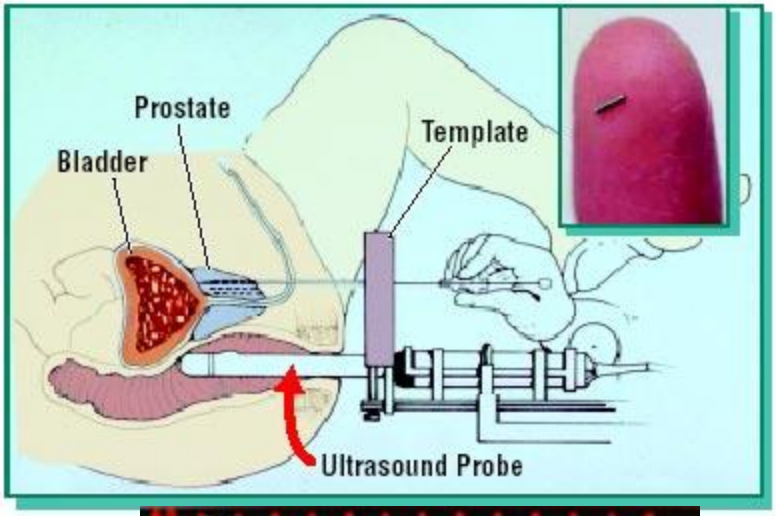


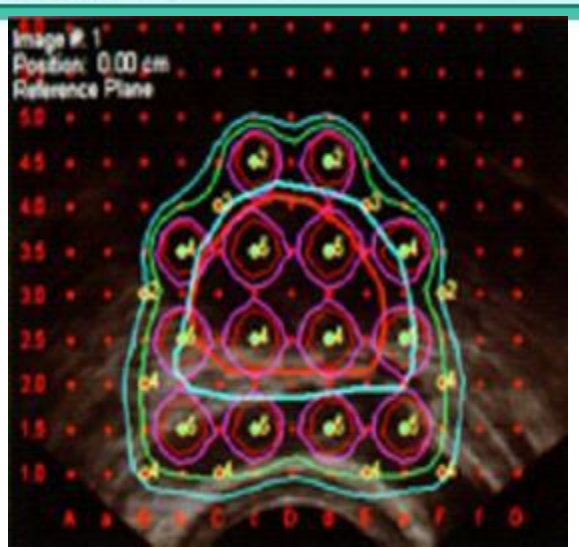




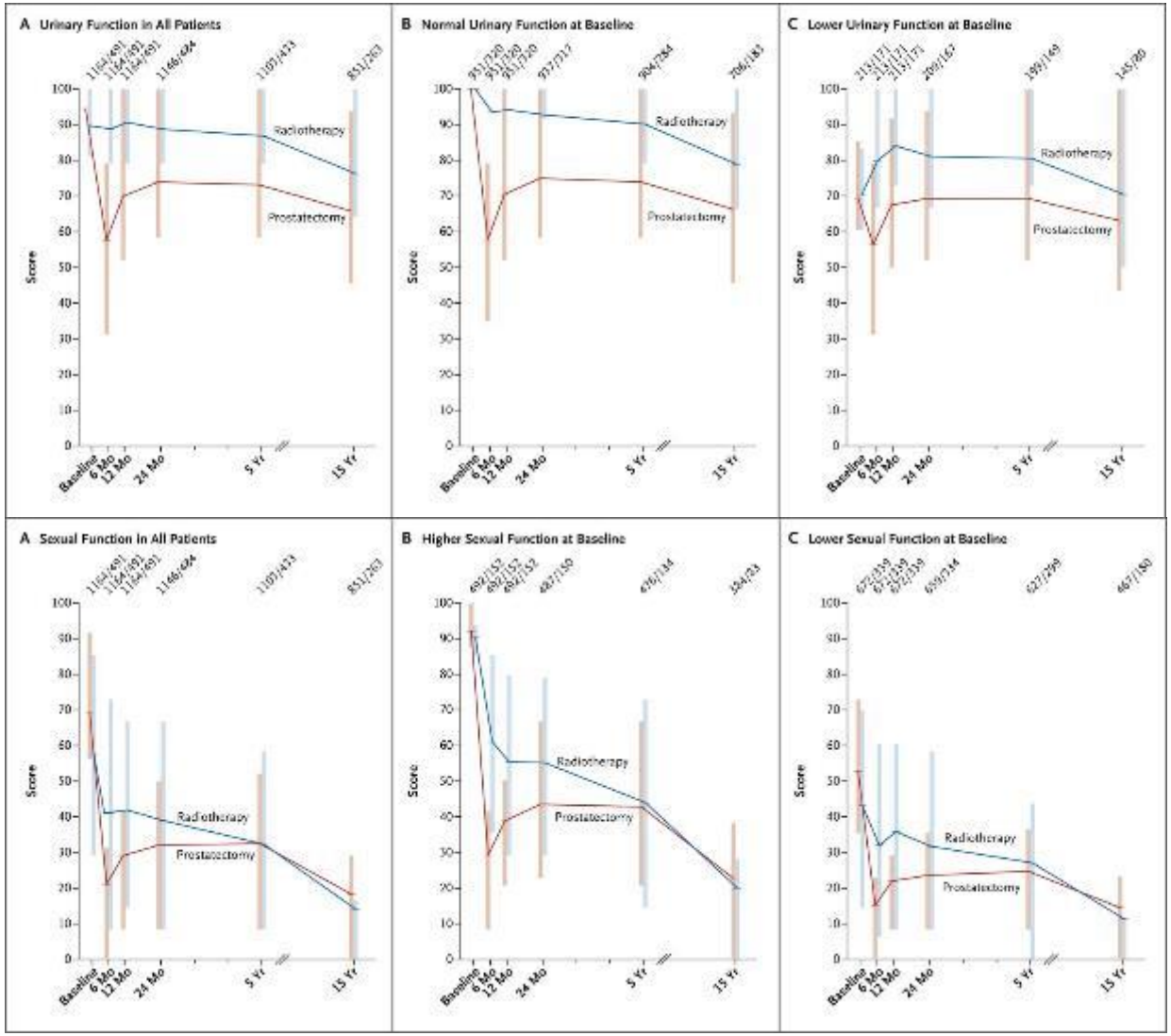
Radiation Therapy:
External beam radiotherapy and Brachytherapy



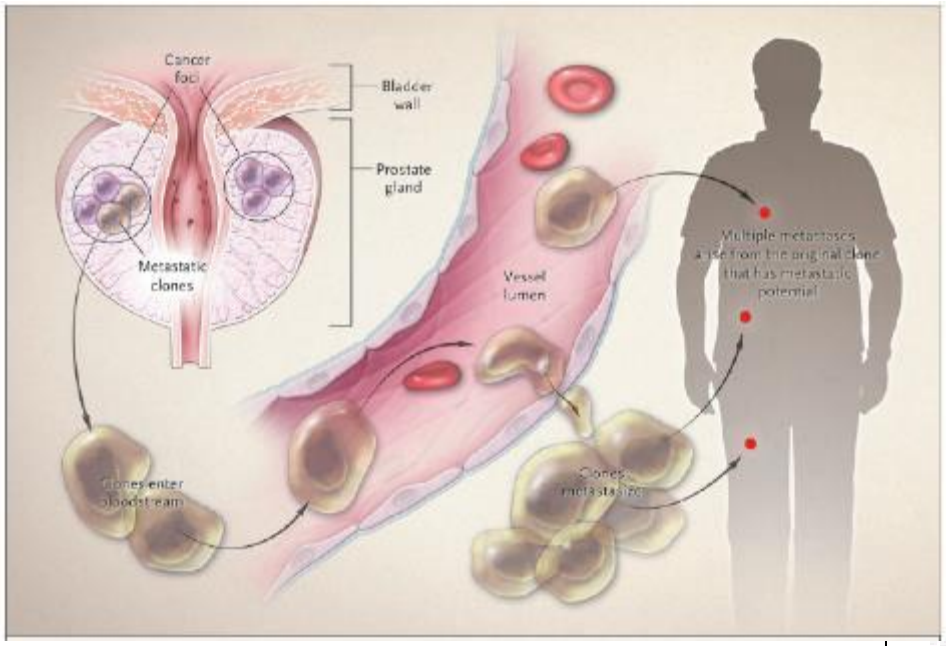


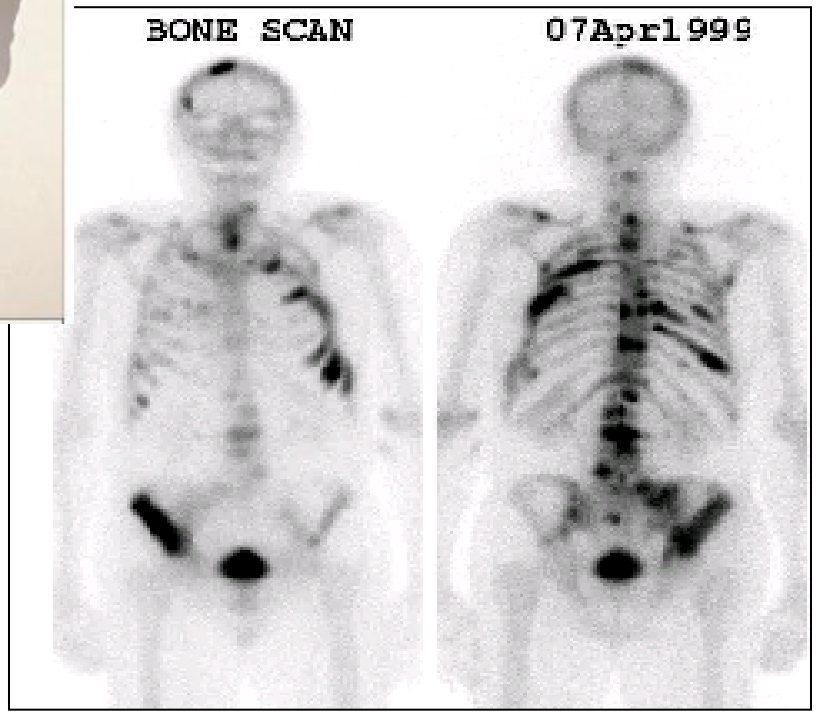


Long-Term Functional Outcomes after Treatment of Localized Prostate Cancer

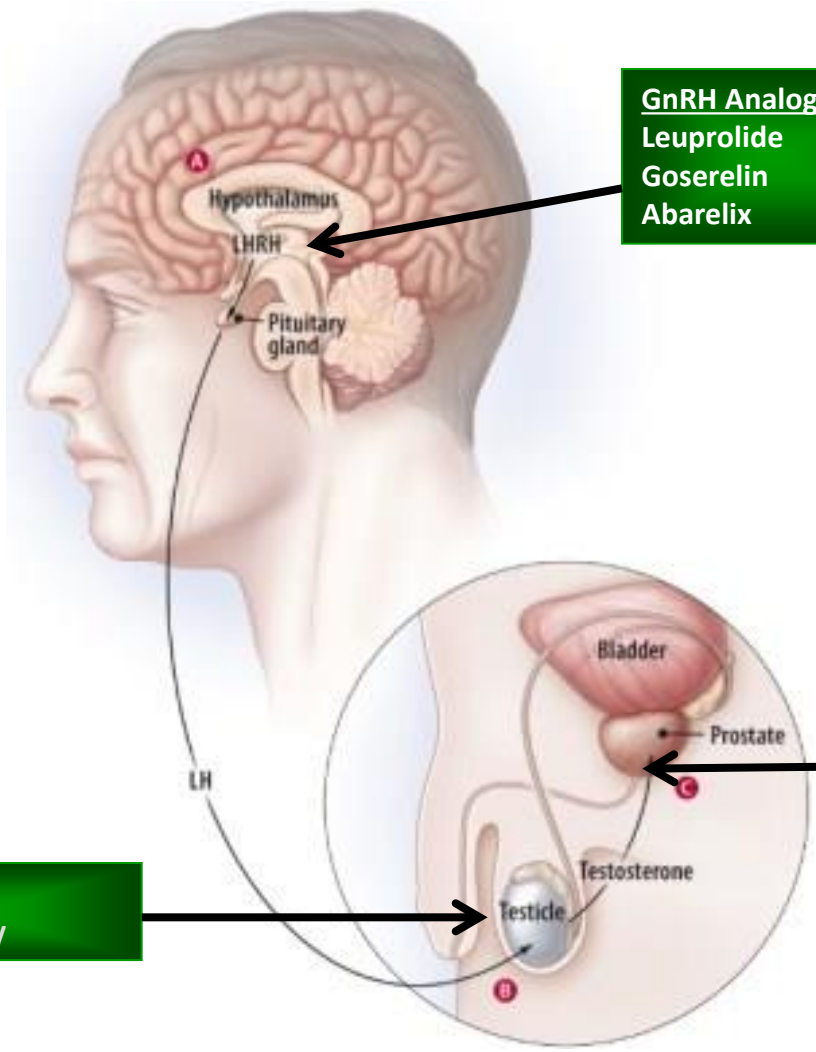


Metastatic Prostate Cancer
Systemic treatment Options





Androgen Deprivation Therapy

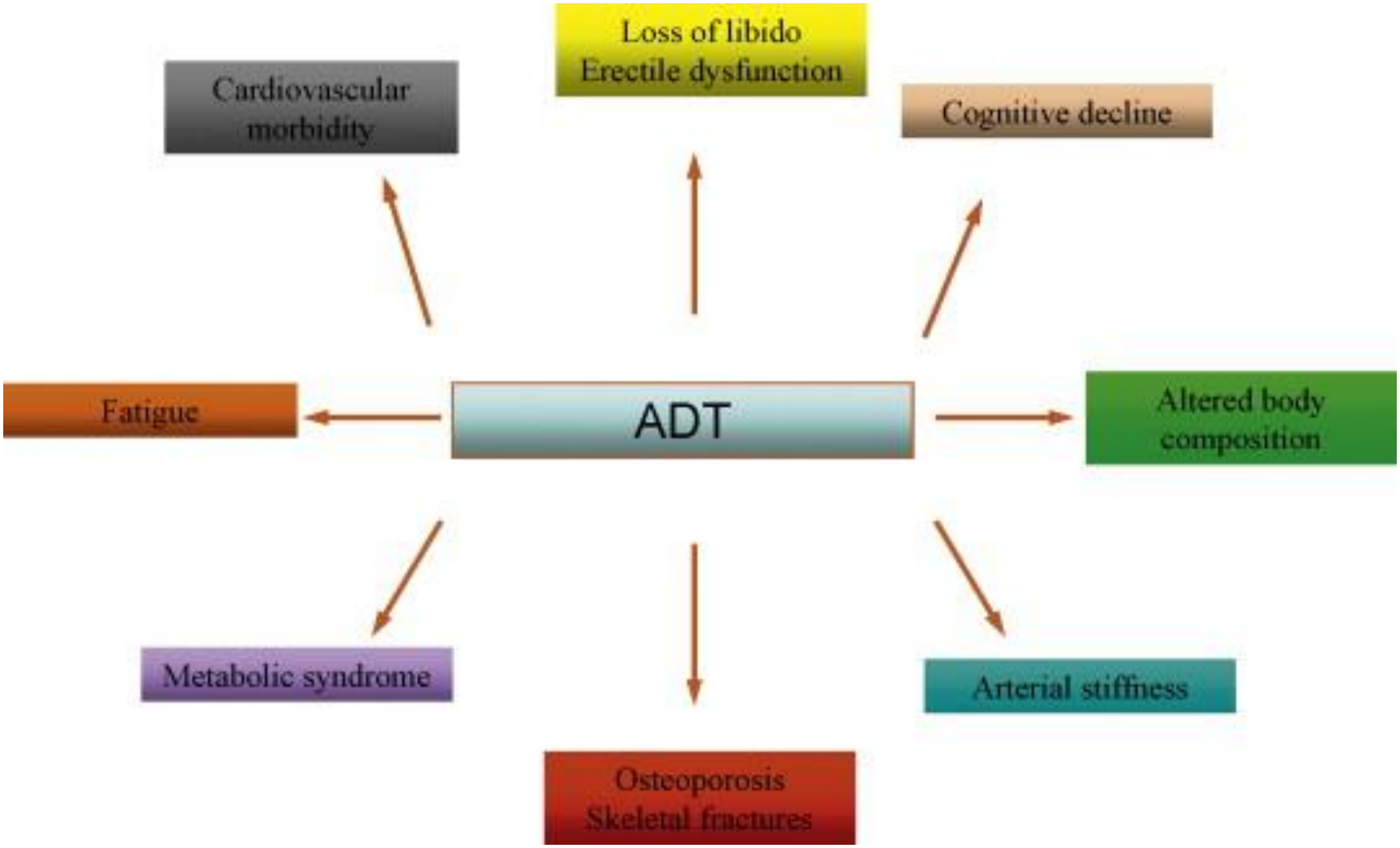


GnRH Analogs
Leuprolide
Goserelin
Abarelix

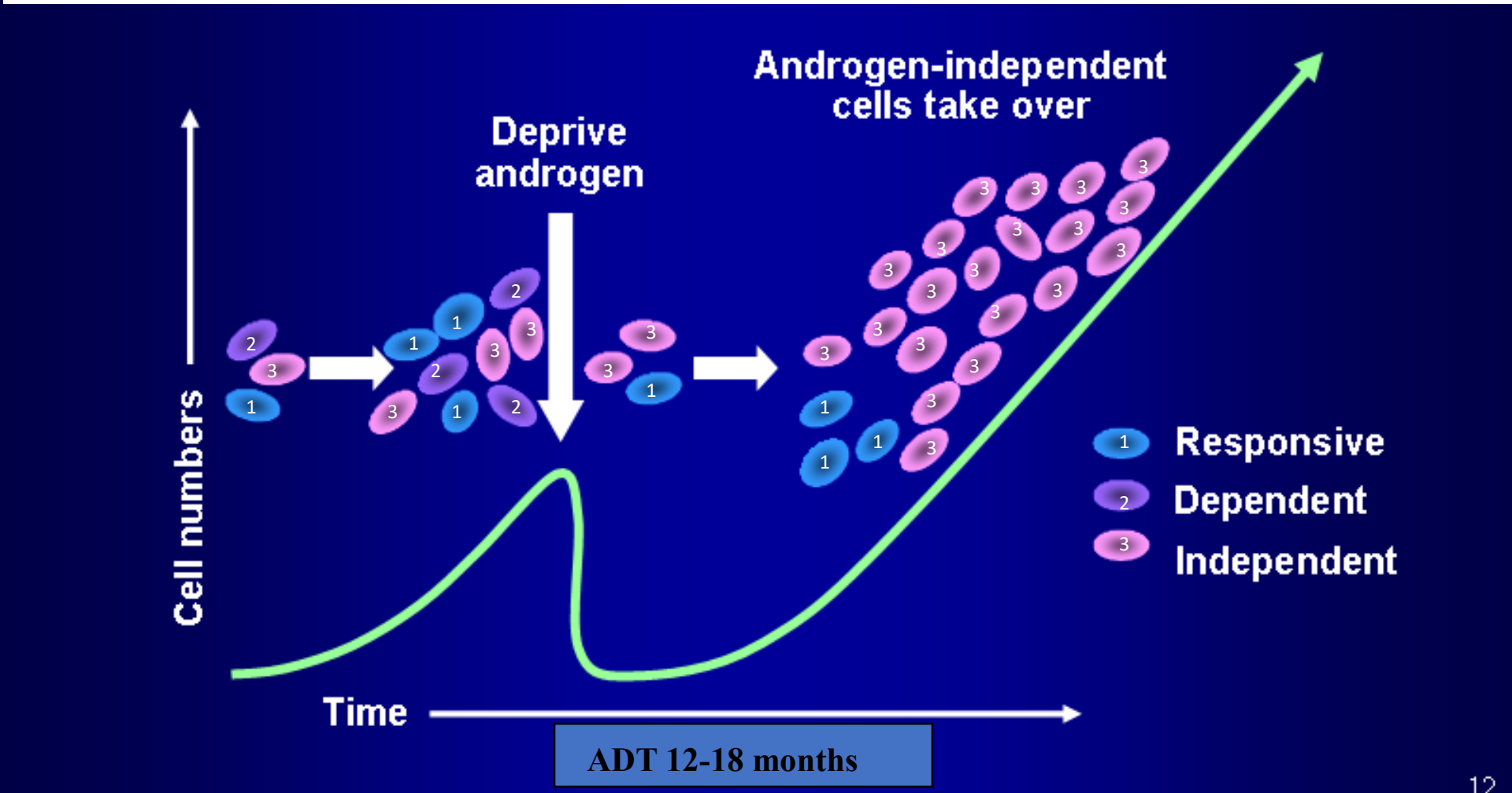
Anti-Androgens
Flutamide
Bicalutamide
Nilutamide

Surgical
Orchiectomy

Androgen Deprivation Therapy – Side Effects



Development of Castrate Resistant Prostate Cancer



Adapted from *Prostate Cancer*. London, England: Times Mirror International Publishers Ltd;1996:143.

Bladder Cancer

Statistics

- About 85,000 new diagnoses annually, 17,000 deaths annually
- 4th most common cancer in men
- 3:1 M:F
- Rare <40
- Most >65 y - medical comorbidities

Table 4. Signs and Symptoms of Bladder Cancer

Hematuria (gross or microscopic)
Irritative symptoms
Dysuria, frequency, urge incontinence, urgency
Obstructive symptoms
Decreased force of stream, feeling of incomplete voiding, intermittent stream, straining
Signs and symptoms of metastases or advanced disease
Abdominal, bone, flank, or pelvic pain; anorexia, cachexia, or pallor; lower extremity edema; renal failure; respiratory symptoms (e.g., cough, dyspnea, hemoptysis); suprapubic palpable mass

Table 1. Risk Factors for Bladder Cancer

Behavioral

Cigarette smoking

Chemical exposure

Arsenic in well water

Chronic infection

Bladder calculi, chronic bladder infection, genitourinary tuberculosis, long-term indwelling catheter, schistosomiasis

Iatrogenic

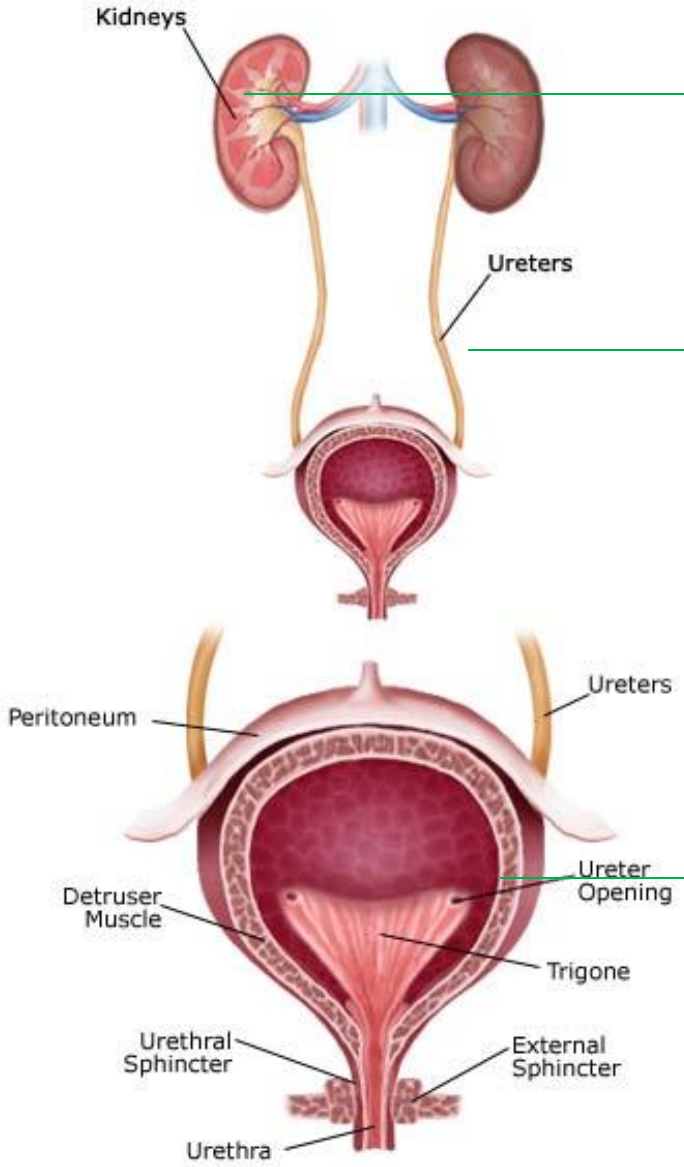
Cyclophosphamide (Cytoxan), pelvic radiation therapy

Occupational

Exposure to aromatic amines (beta-naphthylamine) used in the manufacturing of chemical dyes and pharmaceuticals and in gas treatment plants

Manufacturing (cable, glass, paint, petroleum, tire and rubber, textile); service industry (hairdressers, painters, tar and asphalt workers, truck drivers)

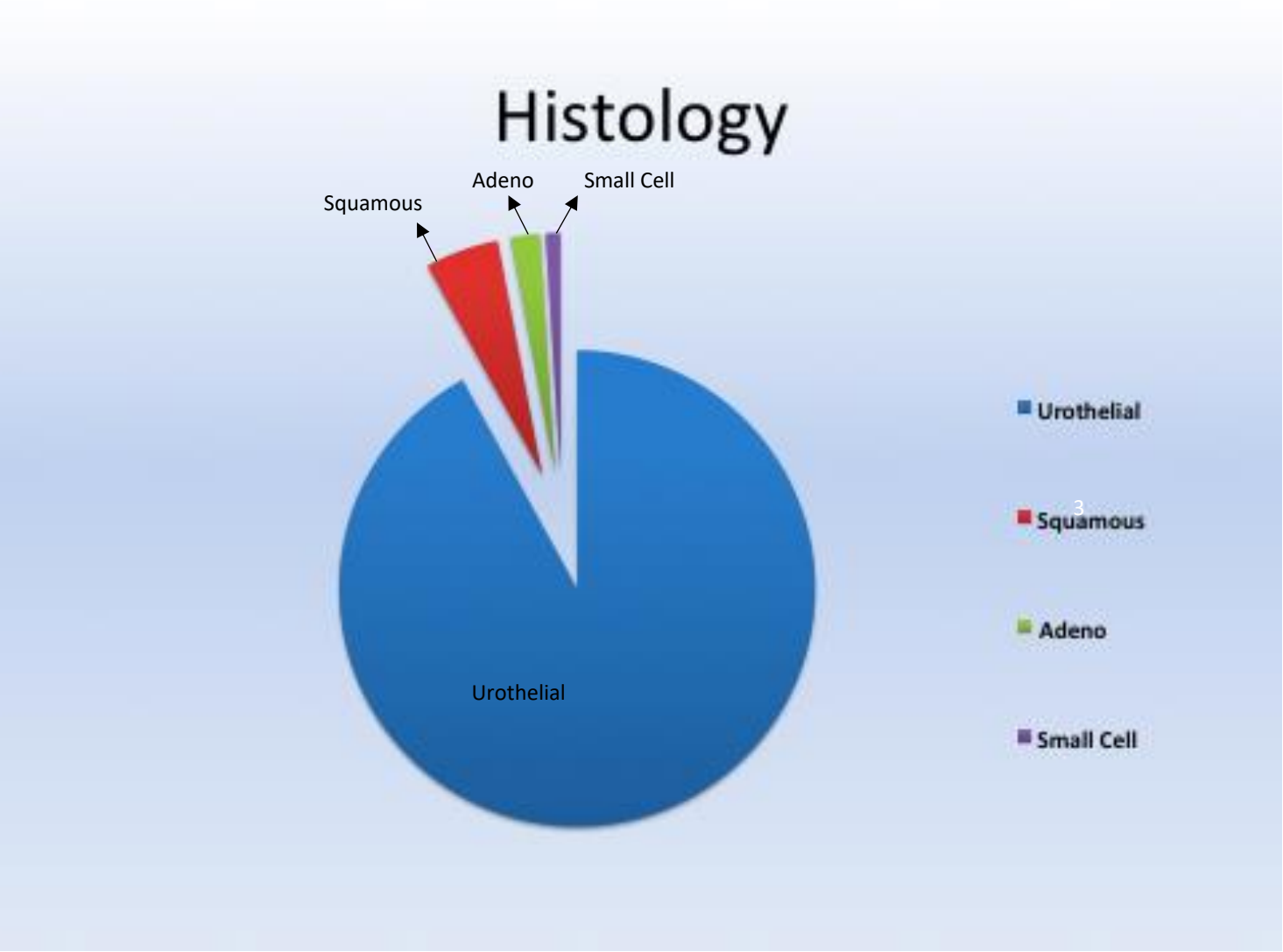
Human Urinary Tract



8%

2%

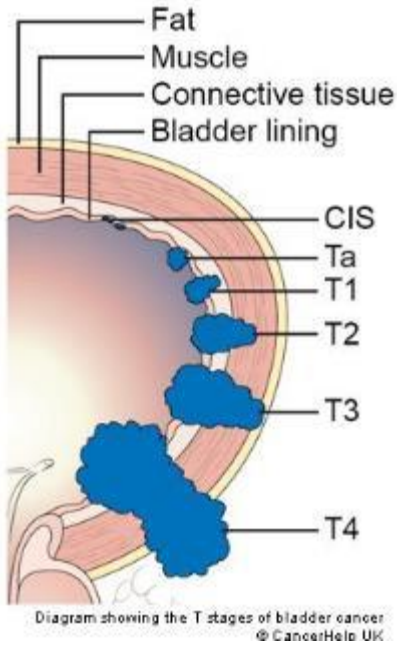
90%



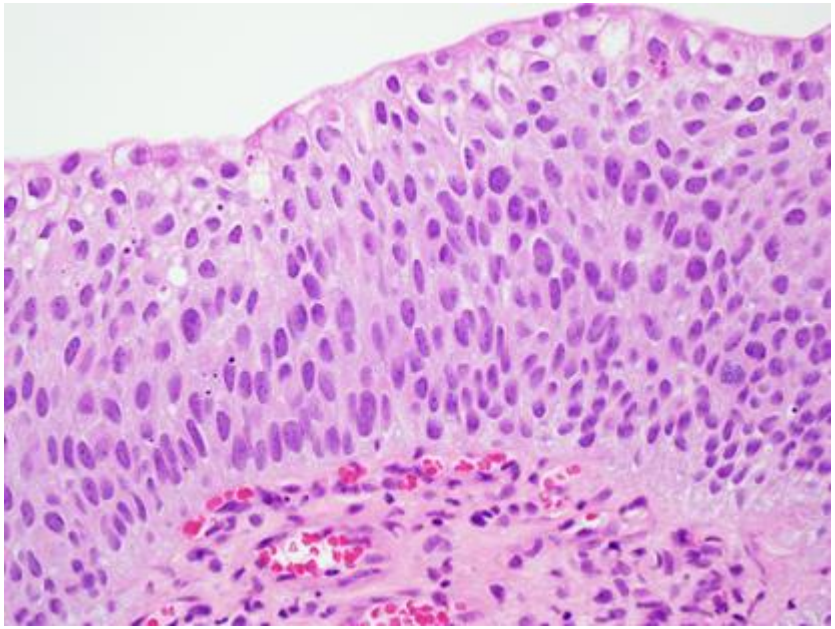
Squamous cell more common in countries where urinary schistosomiasis is common

Urothelial carcinoma

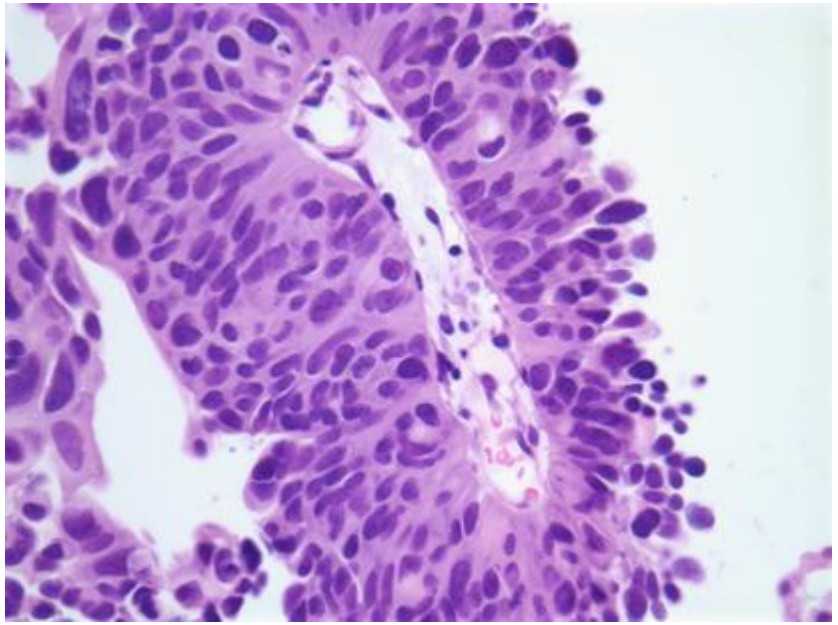
- Low grade non-invasive papillary urothelial carcinoma.
- High grade non-invasive papillary urothelial carcinoma.
- Carcinoma in situ.
- Early invasive urothelial carcinoma (lamina propria invasion).
- Invasive urothelial carcinoma with muscularis propira invasion.



Non invasive papillary urothelial carcinoma

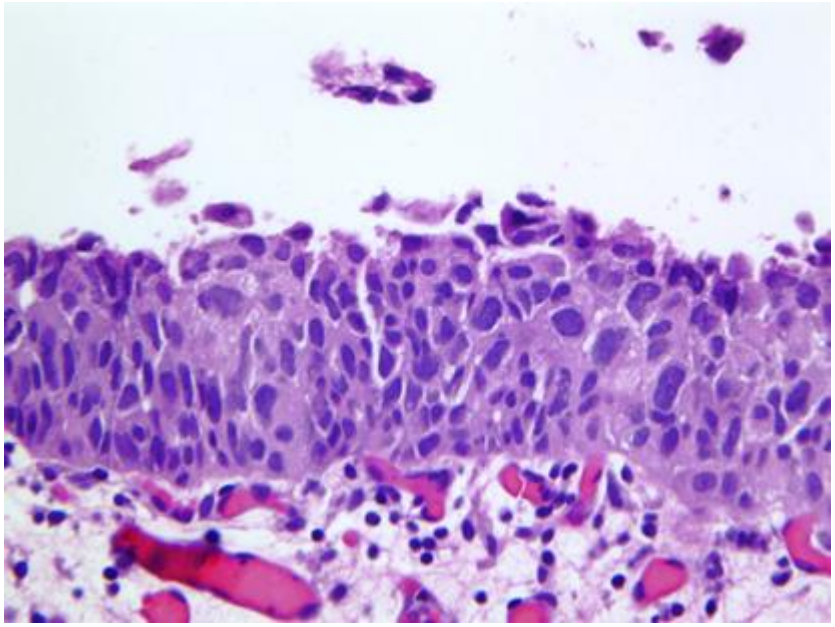


Low-grade papillary urothelial carcinoma. The urothelium is thickened with the architecture generally intact but with some irregularity in the nuclear spacing. There is variability in the nuclear size with variable degrees of hyperchromasia

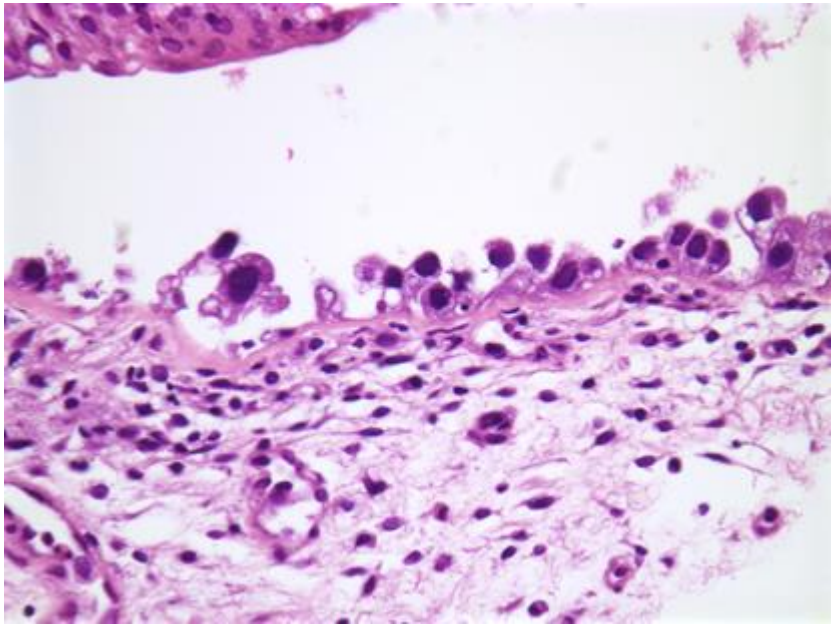


High-grade papillary urothelial carcinoma. There is significant disturbance of the architecture and the cells have pleomorphic and hyperchromatic nuclei. In this example, there is a tendency of the cells to be discohesive.

Urothelial carcinoma in situ

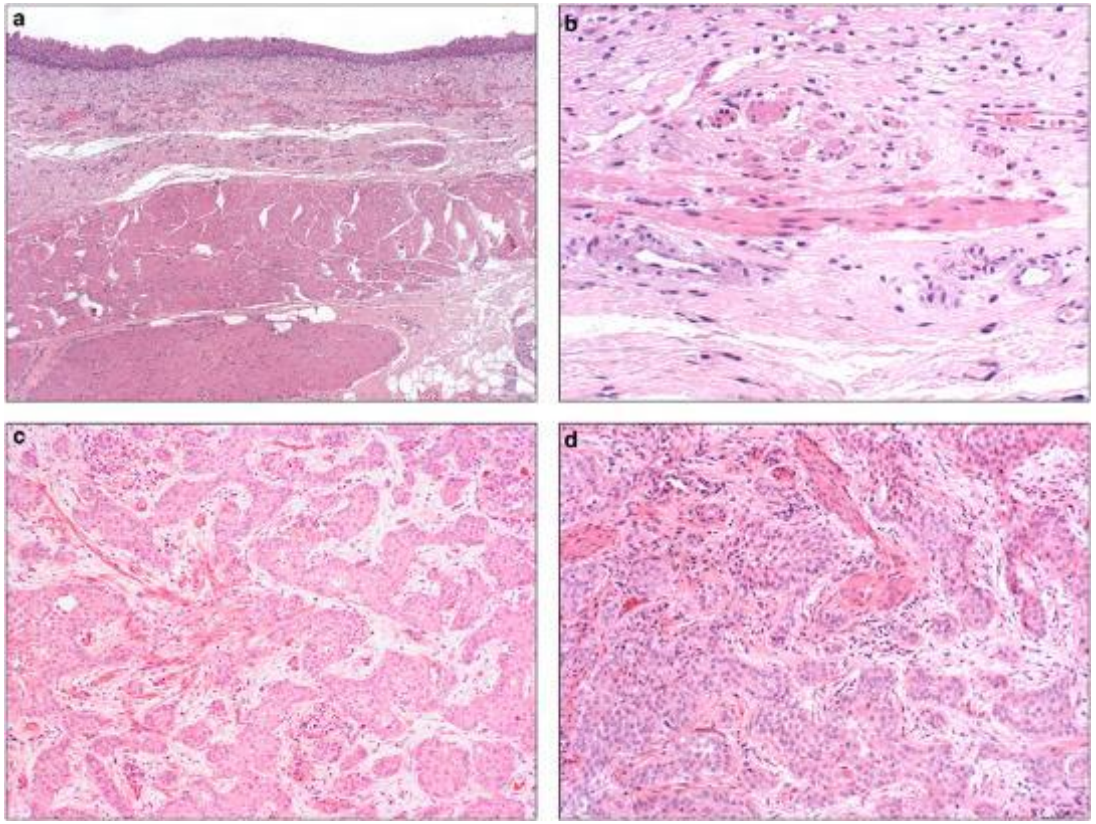


Urothelial carcinoma *in situ*. In this example, there is marked nuclear enlargement and pleomorphism. The architecture is completely disrupted. Note that many of the cells have abundant cytoplasm.



Urothelial carcinoma *in situ*. The denuding pattern is characterized by poorly cohesive cells that typically have large, pleomorphic and hyperchromatic nuclei

Early invasive urothelial carcinoma (pT1)



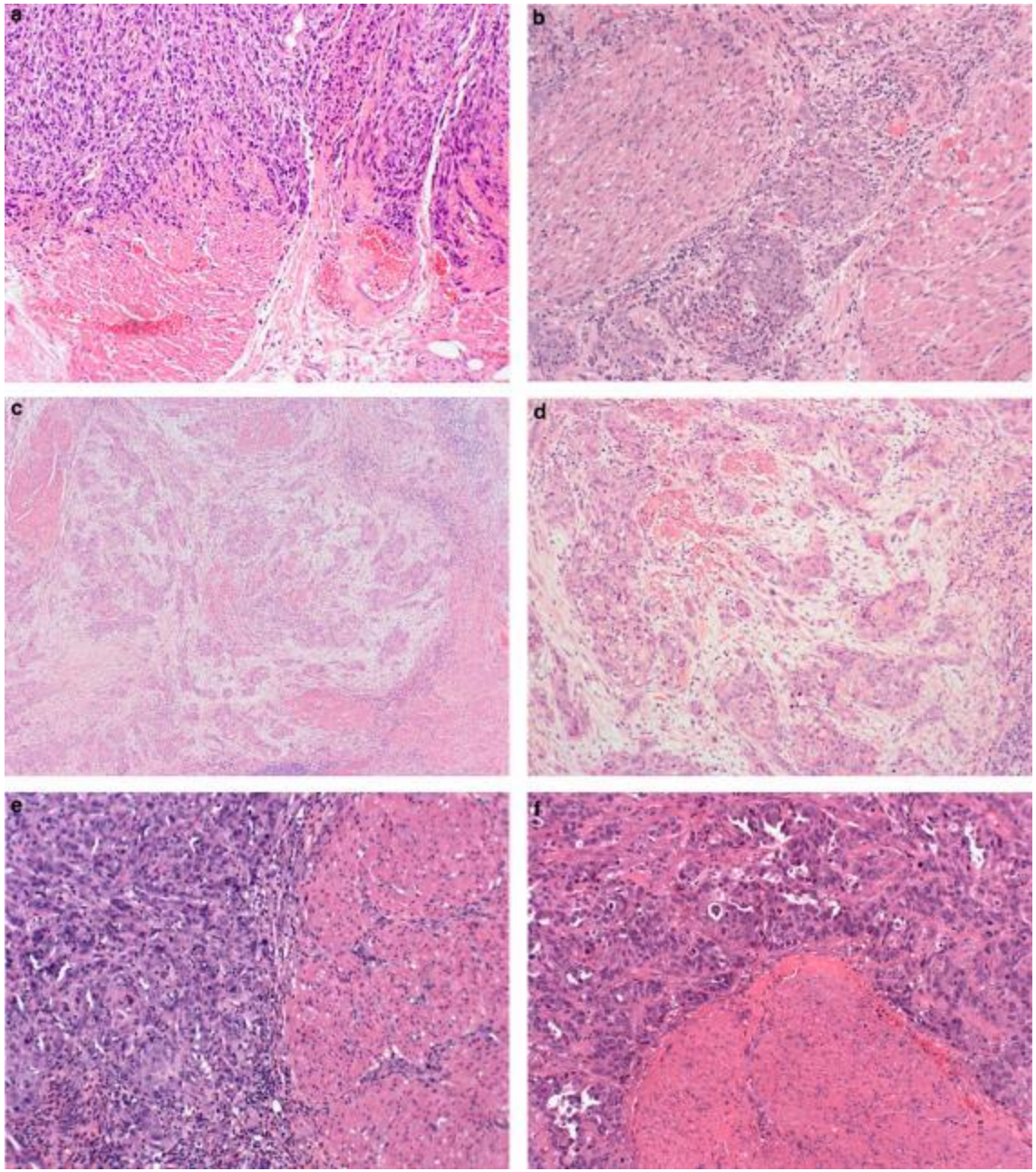
Muscularis mucosae and muscularis mucosae invasion. (**a**) Muscularis mucosae is composed of thin and wavy fascicles of smooth muscle in the submucosa of the bladder wall.

(**b**) Muscularis mucosae is characterized by thin, often interrupted, bands of smooth muscle cells at high power.

(**c**) Muscularis mucosae invasion.

(**d**) Hyperplastic muscularis mucosae may be difficult to be distinguished from muscularis propria.

Muscle invasive urothelial carcinoma (pT2)



(a) pT2 urothelial carcinoma invades into muscularis propria. Infiltrating tumor cells permeate the thick muscularis propria wall.

(b) Sometimes it is difficult to ascertain the presence of muscularis propria invasion when columns and nests of tumor cells widely separate bundles of detrusor muscle.

(c and d) Extensive destruction of muscularis propria may result in fragmentations of detrusor muscle, mimicking muscularis mucosae invasion.

(e) It is not uncommon for tumor cells to abut, but not penetrate bundles of mucularis propria. The muscle–tumor interface may even have a smooth contour.

(f) Even without direct invading into muscularis propria wall, tumor cells immediately adjacent to broad smooth muscle fibers should be categorized as pT2 carcinoma

Staging/management

Categories

- Non-Invasive

  prevent recurrence and progression
- Invasive

  preserve bladder or prevent progression
- Metastatic

  maintain quality and prolong life

Table 3. American Joint Committee on Cancer TNM Staging System for Bladder Cancer

Primary tumor (T)

TX: Primary tumor cannot be assessed

T0: No evidence of primary tumor

Ta: Noninvasive papillary carcinoma

Tis: Carcinoma in situ ("flat tumor")

T1: Tumor invades subepithelial connective tissue

T2: Tumor invades muscle

pT2a: Tumor invades superficial muscle (inner half)

pT2b: Tumor invades deep muscle (outer half)

T3: Tumor invades perivesical tissue

pT3a: Microscopically

pT3b: Macroscopically (extravesical mass)

T4: Tumor invades any of the following: prostate, uterus, vagina, pelvic wall, abdominal wall

T4a: Tumor invades prostate, uterus, vagina

T4b: Tumor invades pelvic wall, abdominal wall

Regional lymph nodes (N)

Regional lymph nodes are those within the true pelvis; all others are distant lymph nodes

NX: Regional lymph nodes cannot be assessed

N0: No regional lymph node metastasis

N1: Metastasis in a single lymph node, 2 cm or less in greatest dimension

N2: Metastasis in single lymph node, more than 2 cm but not more than 5 cm in greatest dimension; or multiple lymph nodes, none more than 5 cm in greatest dimension

N3: Metastasis in lymph node more than 5 cm in greatest dimension

Distant metastasis (M)

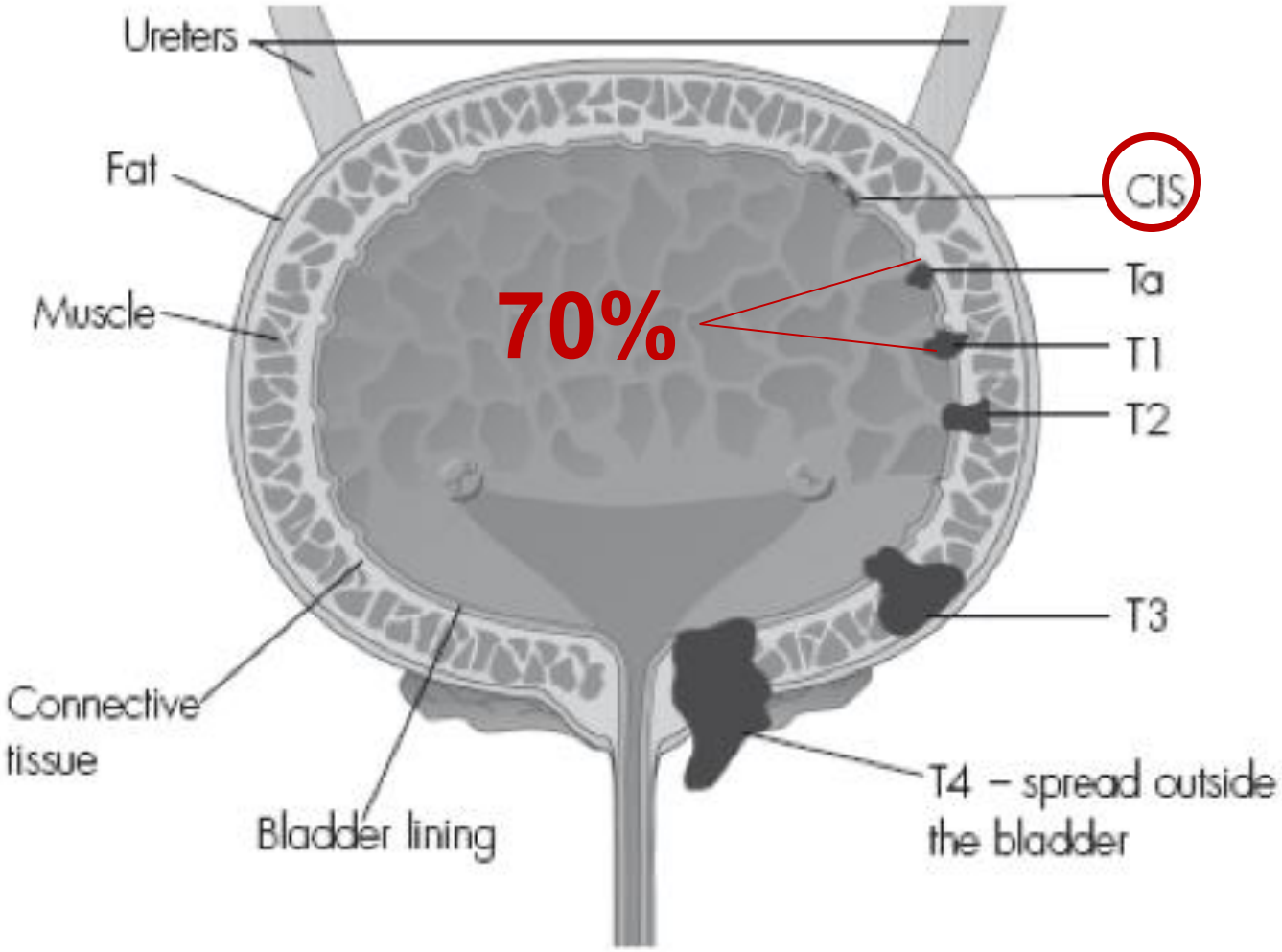
MX: Distant metastasis cannot be assessed

M0: No distant metastasis

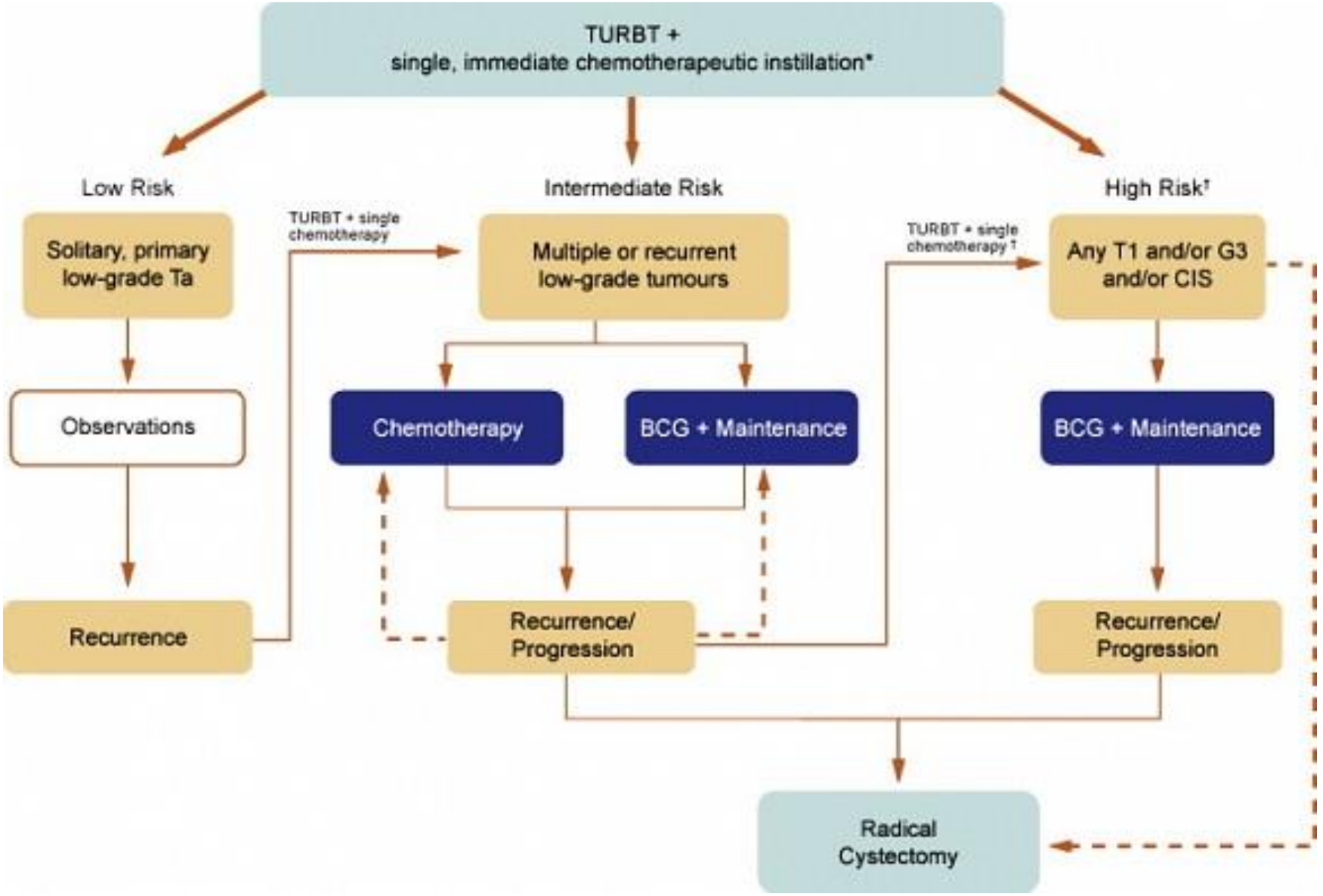
M1: Distant metastasis

Stage grouping

0a	Ta	N0	M0
0is	Tis	N0	M0
I	T1	N0	M0
II	T2a	N0	M0
	T2b	N0	M0
III	T3a	N0	M0
	T3b	N0	M0
	T4a	N0	M0
IV	T4b	N0	M0
	Any T	N1	M0
	Any T	N2	M0
	Any T	N3	M0
	Any T	Any N	M1


Ureters
Fat
Muscle
Connective tissue
Bladder lining
70%
CIS
Ta
T1
T2
T3
T4 – spread outside the bladder

Treatment of Bladder Cancer
Early Stage and Muscle Invasive Disease



Eur Urol Volume 56, issue 3, pages 407-582, September 2009

Complications of Intravesical Therapy

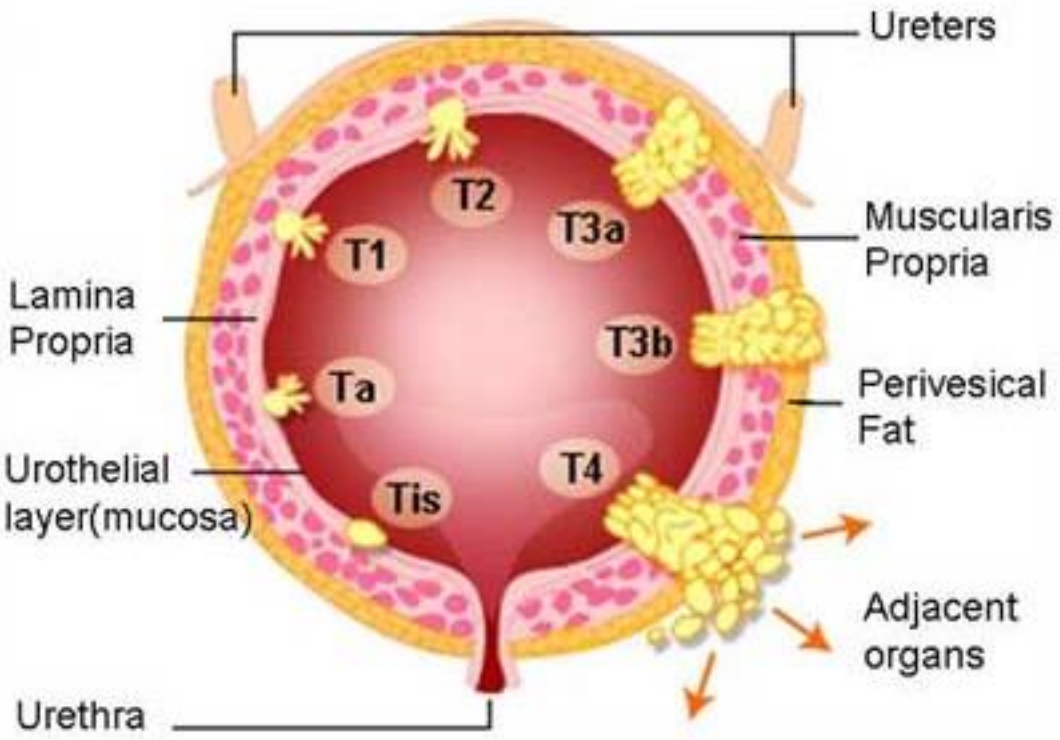
Drug	Side Effects	Patients Affected (%)
Bacille Calmette-Guérin	Cystitis (noninfectious) Fever (low grade), chills, malaise Symptomatic prostatitis Epididymitis Fever (≥ 103°F) Sepsis Distant organ (ie, pulmonary miliary pattern)	> 50 25 < 5 < 2 3 < 0.5 Rare
Mitomycin C	Chemical cystitis Myelosuppression Rash (hypersensitivity dermatitis) Bladder contracture Tissue necrosis	10 < 10 < 10 < 2 Rare
Thiotepa	Chemical cystitis Myelosuppression	20 10
Doxorubicin	Chemical cystitis Allergic reaction	25 1

Sources:

Hall MC, Chang SS, Dalbagni G, et al. Guideline for the management of nonmuscle invasive bladder cancer (stages Ta, T1, and Tis): 2007 update. *J Urol.* 2007;178(6):2314-2330.

Lamm DL, McGee WR, Hale K. Bladder cancer: current optimal intravesical treatment. *Urol Nurs.* 2005;25(5): 323-326, 331-332.

Muscle Invasive Bladder Cancer



TURBT

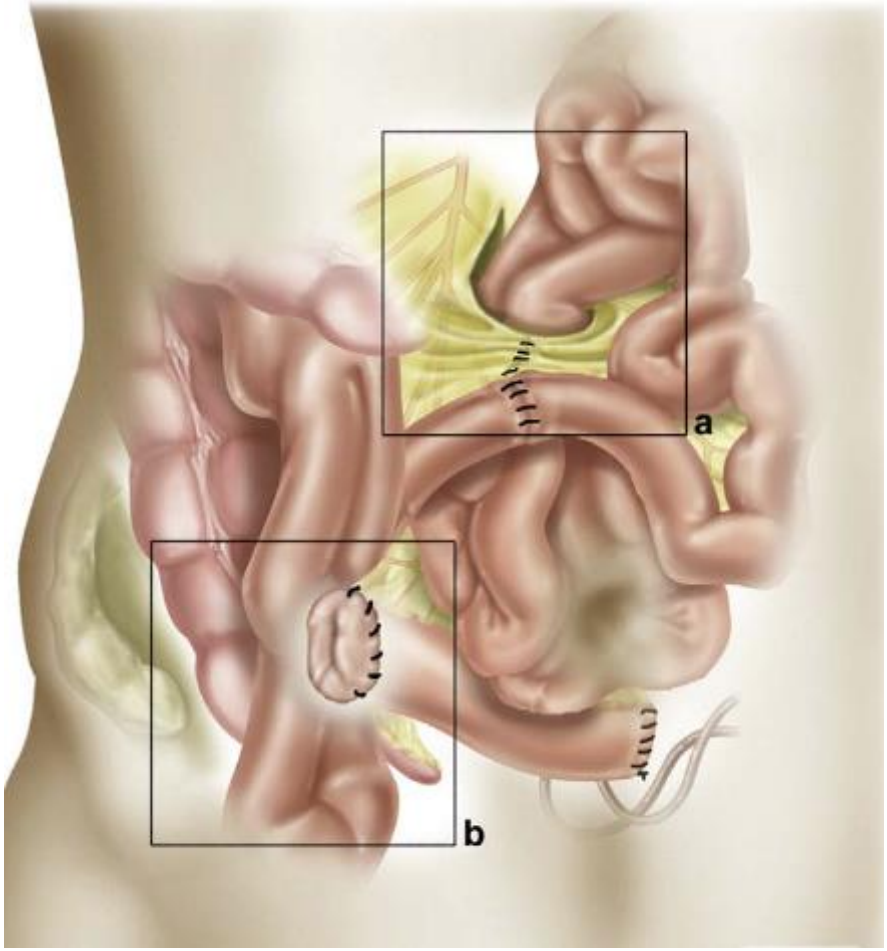
Staging to R/O Metastasis

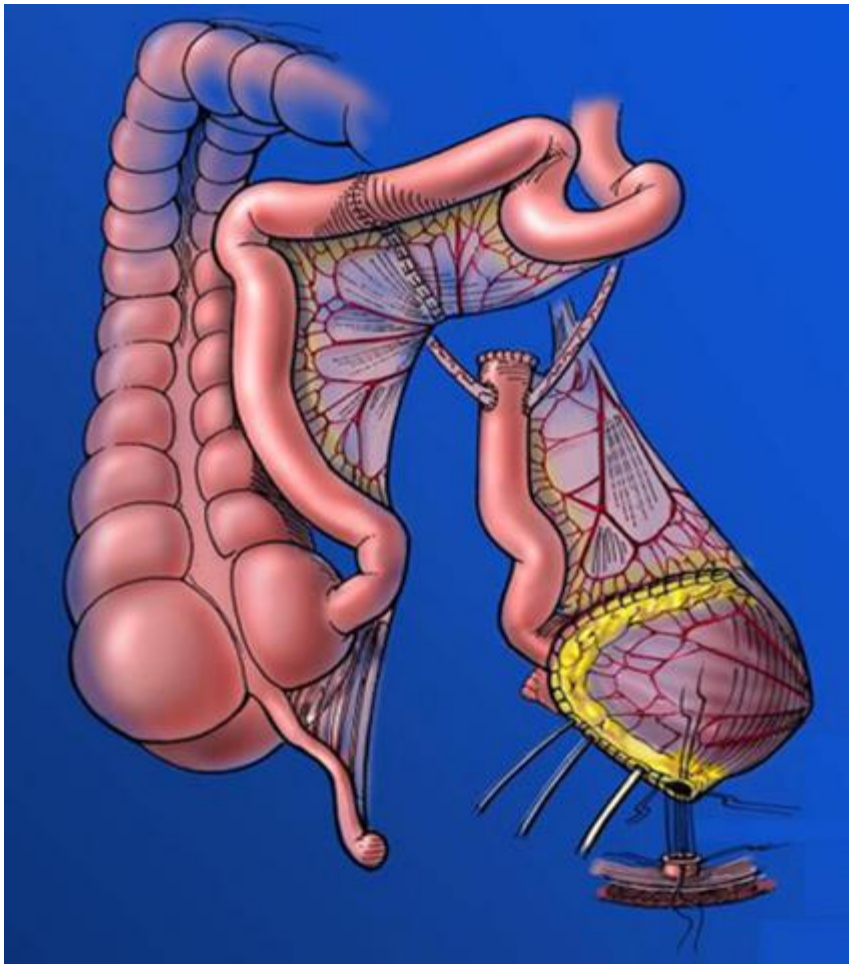
Chemotherapy

Consolidation

Urinary Diversions – and metabolic considerations

Ileal Conduit





EUROPEAN UROLOGY SUPPLEMENTS 9 (2010) 736–744

Summary

- Bladder cancer account for ~7% of all cancers and ~3% of all cancer deaths in the US (vast majority are urothelial)
- Main types:
  1. High and Low grade non-invasive papillary urothelial carcinomas.
  2. Urothelial carcinoma in situ.
  3. Invasive urothelial carcinomas (early and advanced).
     - The extent of invasion and spread (staging) at the time of initial diagnosis is the most important prognostic factor
     - Radical cystectomy is typically reserved for: T2 tumors, CIS or HG refractory to BCG/recurrent or CIS extending into prostatic urethra and down prostatic ducts.

Testis Cancer









Epidemiology

- 2-3 new cases per 100,000 US people with testes per year
- Marked variation in incidence among different countries/races
- 90-95% are germ cell
- Most common solid tumor in people with testes ages 15-35

Risk Factors

- Cryptorchidism: 7-10% of patients with testicular cancer have a history of cryptorchidism
    - Abnormal germ cell morphology
    - Elevated temperature
    - Interference with normal blood supply
  - 5-10% of patients with testicular cancer and a history of cryptorchidism develop cancer in the contralateral testis
  - Orchidopexy does not prevent development of cancer – just allows for detection

Risk Factors

- Gonadal Dysgenesis
  - 20-30% develop cancer (gonadoblastoma)
- Trauma ??
  - prompts evaluation
- Hormones
  - DES/OCP probably do not increase risk
- Atrophy (nonspecific vs. mumps orchitis)
  - Speculative

Pathology

- Most (~95%) of testicular cancer in postpubertal people are germ cell tumors
- Arise from intratubular germ cell neoplasia (common in cryptorchidism and dysgenic gonads
- Subclassified as seminomas and non-seminomatous germ cell tumors.
- 50% are classic seminoma (histologically identical to ovarian and CNS dysgerminomas).
- Other types include embryonal, yolk sac, choriocarcinoma and teratoma.
- Mixed germ cell tumors contain more than 1 type usually embryonal, teratoma, yolk sac.

Pathology

GERM CELL

- Seminoma 30-60%
- Embryonal 3-4%
- Yolk sac
- Teratoma 5-10%
- Choriocarcinoma 1%
- Mixed 40%

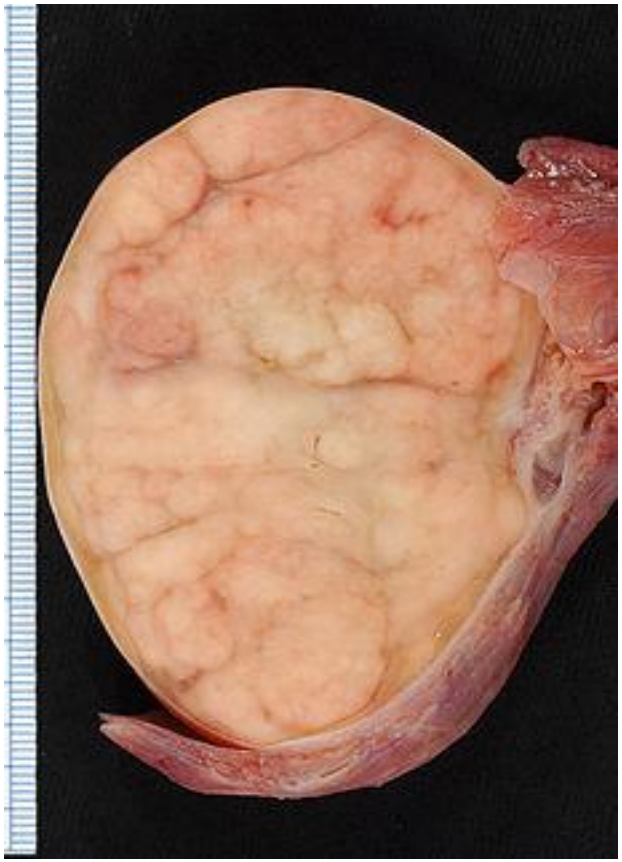
NONGERM CELL

- Leydig 1-3%
- Sertoli <1%
- Gonadoblastoma 0.5%

Tumor	Peak age	Morphology	Tumor makers
Seminoma	40-50	Sheets of uniform polygonal cells with clear cytoplsm and stromal lymphocytes	10% HCG
Embryonal	20-30	Poorly diff pleomorphic cell in cords, sheets or papillary formations; can contain yolk sac and chroio	Neg (pure)
Yolk sac	3	Poorly diff endothelium-like, cuboida or columnar	90% AFP
Choriocarcinoma	20-30	Cytotrophoblas and syncytiotrophoblasts without villus formation	100% HCG
Teratoma	All	Tissue from all three germ cell layers with varying degrees of diff	Neg (pure)
Mixed	15-30	Variable	90% HCG and AFP

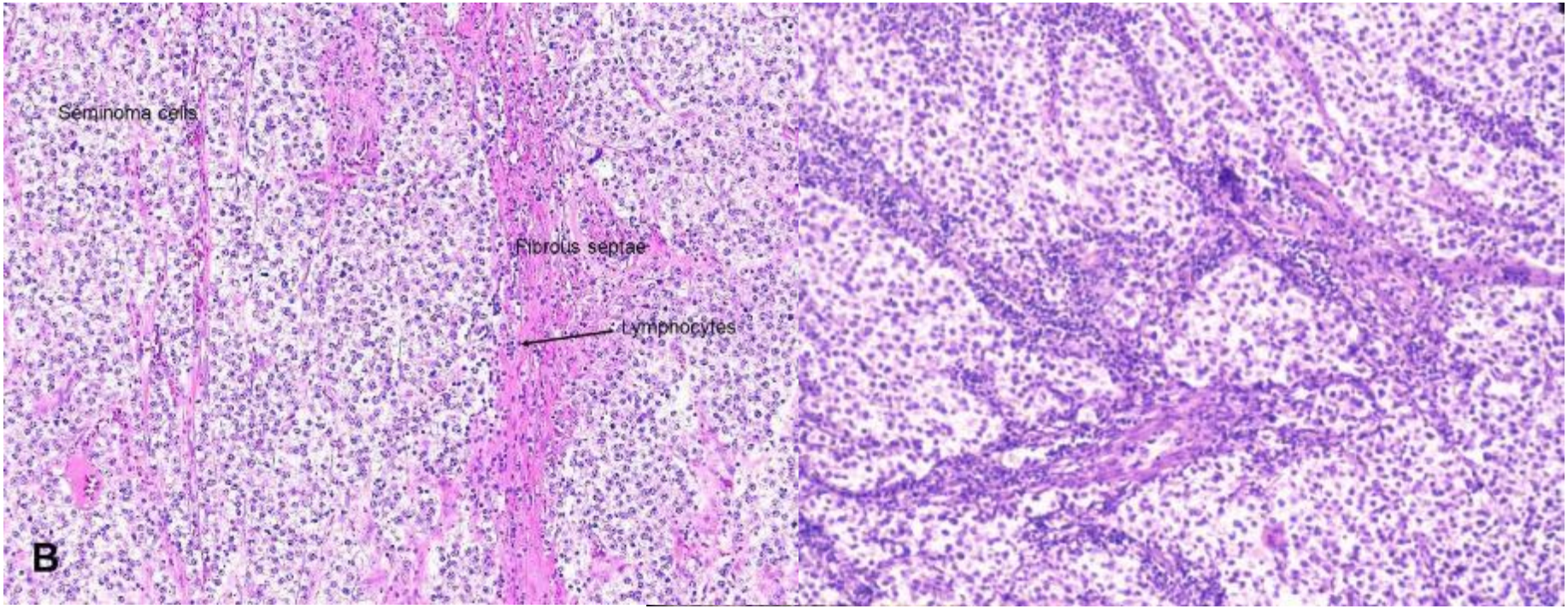
Seminoma

- Peak age 40-50 years
- 10% have elevated HCG
- Soft, well demarcated, grey-white tumors, bulging from cut surface
- Larger tumors may have areas of necrosis, usually without hemorrhage.



Seminoma

- Sheets of uniform polygonal cells with cleared cytoplasm and stromal lymphocytes.
- Round nuclei with prominent nucleoli.
- Often arranged in small lobules with fibrous septa
- 15% have syncytiotrophoblasts (source of HCG elevation).
- Although similar in name spermatocytic seminomas are distinct unrelated tumors.



- Symptoms/work-up/treatment

Symptoms

- **Painless swelling / mass**
- **30-40% dull / aching sensation**
- **10% present with metastatic symptoms**
- **Gynecomastia**
  - **5% germ cell**
  - **30-50% Sertoli/Leydig**
- **1-2% have bilateral disease at diagnosis**
- **More common on the right**
- **Infertility**

Work-up

- **Exam**
- **U/S**
- **CXR +/- Chest CT**
- **Abdominal CT**
  - **Can identify small nodal deposits <2 cm**
  - **MRI and PET scan no advantage over CT**
- **Serum markers: AFP, BHCG and LDH**
  - **Elevation after orchiectomy represents metastatic disease**
  - **Conversely normalization does not rule out metastatic disease**

Fertility preservation – Sperm Banking

Stage

- Based on:
  1. Extent of tumor invasion (pT)
  2. Regional lymph node metastases (pN)
  3. Distant metastasis (pM)
  4. Serum markers (S)

	Primary tumor (T)
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Intratubular germ cell neoplasia
T1	Tumor limited to the testis and epididymis or tumor invasion into the tunica albuginea only
T2	Tumor extending through the tunica albuginea with involvement of the tunica vaginalis
T3	Tumor invades the spermatic cord
T4	Tumor invades the scrotum
	Regional lymph nodes — clinical (N) or pathologic (pN) staging
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastases to single or multiple lymph nodes, each < 2 cm in size
N2	Metastases to single or multiple lymph nodes, >2 cm but < 5 cm in size
N3	Metastases to lymph node, >5 cm in greatest dimension
	Distant metastasis (M)
MX	Distant metastasis cannot be assessed
M0	No distant metastasis
M1	Distant metastasis
M1a	Nonregional nodal or pulmonary metastasis
M1b	Distant metastasis other than to nonregional lymph nodes and lungs

	tumor markers
S0	Normal level
SX	Unavailable or not performed
S1	Lactate dehydrogenase (LDH) level < 1.5 times normal, human chorionic gonadotropin (HCG) level < 5000 IU/L, alpha-fetoprotein (AFP) level < 1000 ng/mL
S2	LDH 1.5–10 times normal; HCG level, 5000–50,000 IU/L; AFP level, 1000–10,000 ng/mL
S3	LDH >10 times normal; HCG level >50,000 IU/L; AFP level >10,000 ng/mL

Alpha-Fetoprotein

- Expressed by the early embryo (also liver and GI tract)
- Half-life: 5-7 days
- Produced by pure embryonal, teratocarcinoma, yolk sac, mixed tumors (NOT pure choriocarcinoma or seminoma)
- Falsely elevated in liver dysfunction, viral hepatitis and ETOH
- Presence of AFP in what's thought to be pure seminoma suggests a mixed tumor

Human Chorionic Gonadotrophin

- Secretory product of the placenta
- Alpha unit (LDH,FSH,TSH) and beta unit
- Half-life: 24-36 hours
- Produced by syncytiotrophoblastic tissue
- All choriocarcinomas, 40-60% embryonal, 5-10% seminoma
- Falsely elevated in hypogonadism and marijuana use

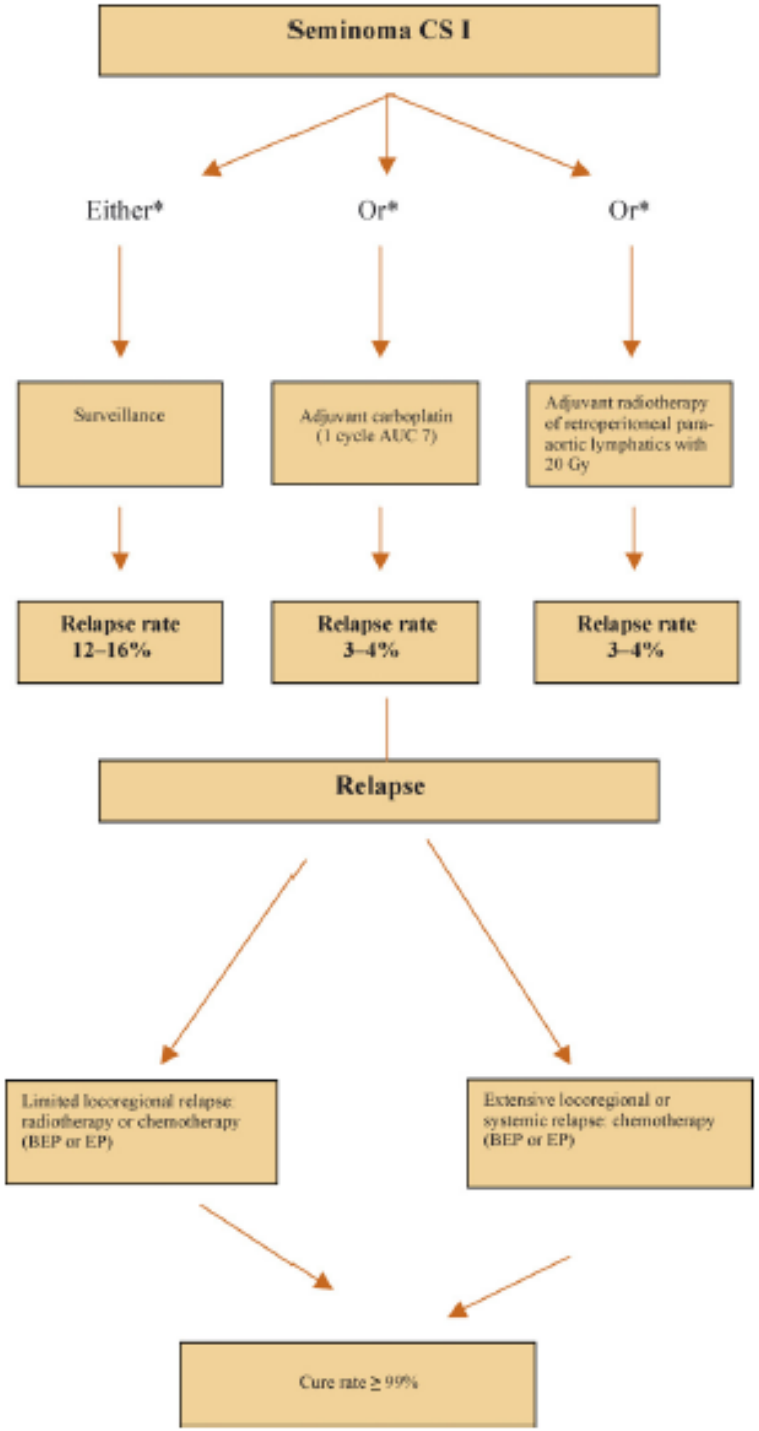
LDH

- **Lactic Acid Dehydrogenase**
  - **Presents normally in smooth, cardiac and skeletal muscle, liver and brain**
  - **Most useful in advanced seminoma or tumors where other markers are not elevated**
  - **Many false positives**

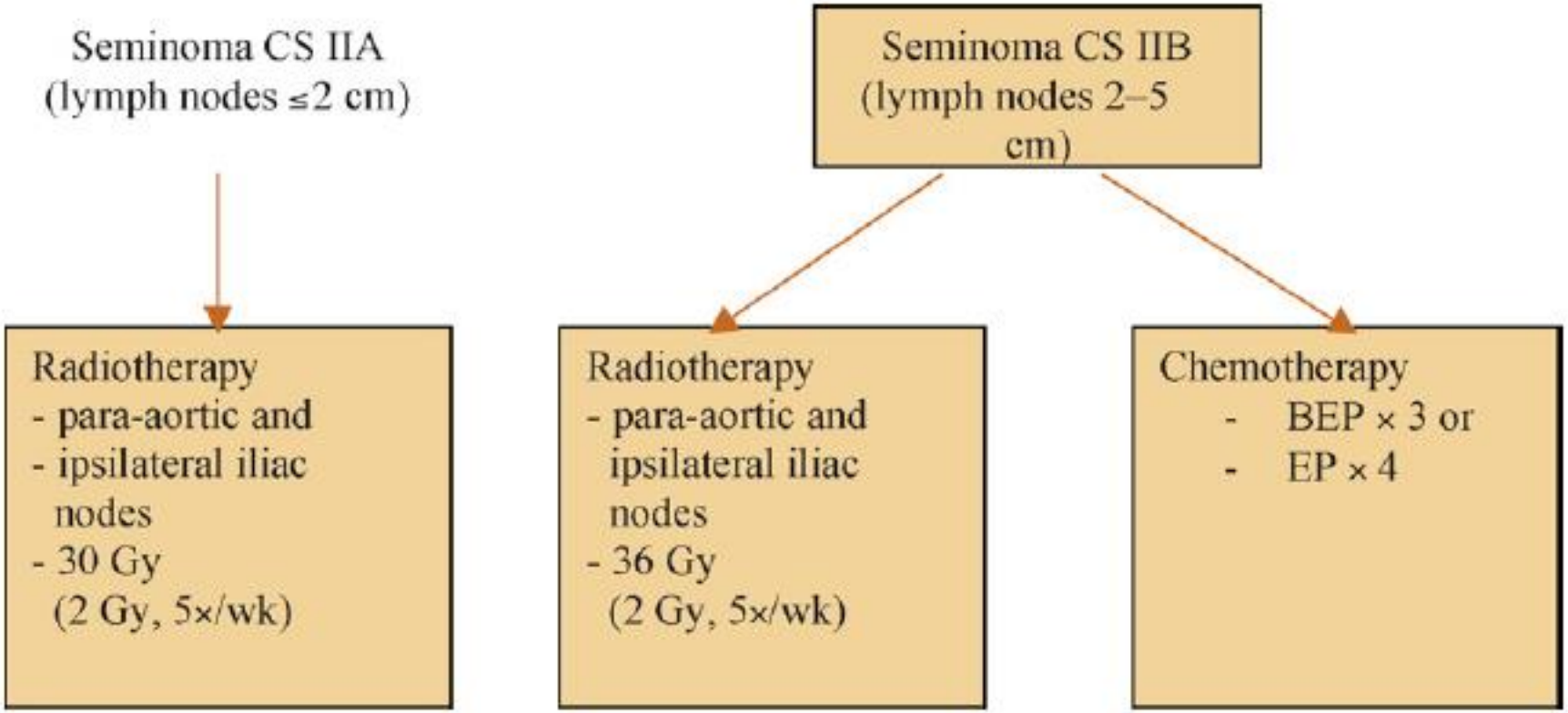
Radical Orchiectomy



- Inguinal approach
- Avoid seeding the scrotum and disrupting lymphatics
- Wait 5 half lives before rechecking markers
- Biopsy not recommended due to the risk of tumor spread



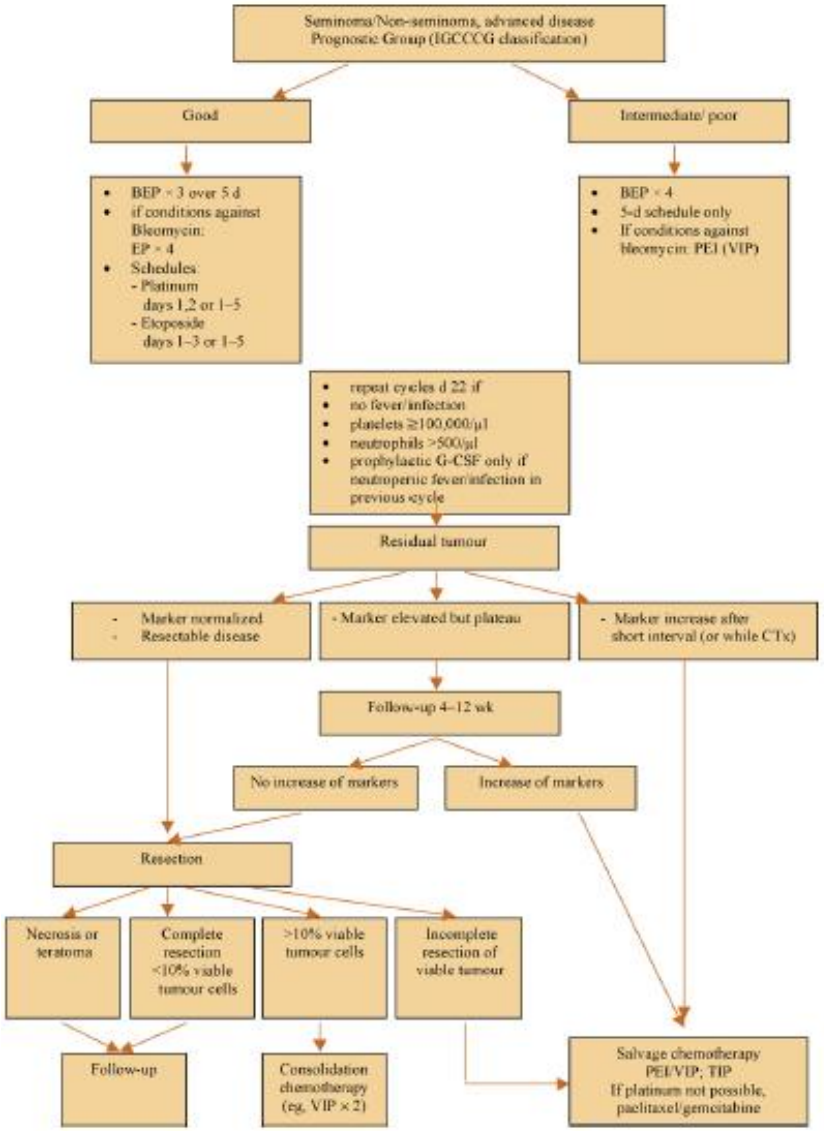
european urology
53 (2 0 0 8) 478–496


Seminoma CS IIA
(lymph nodes ≤2 cm)
Seminoma CS IIB
(lymph nodes 2–5
cm)
Radiotherapy
- para-aortic and
- ipsilateral iliac
nodes
- 30 Gy
(2 Gy, 5×/wk)
Radiotherapy
- para-aortic and
ipsilateral iliac
nodes
- 36 Gy
(2 Gy, 5×/wk)
Chemotherapy
- BEP × 3 or
- EP × 4

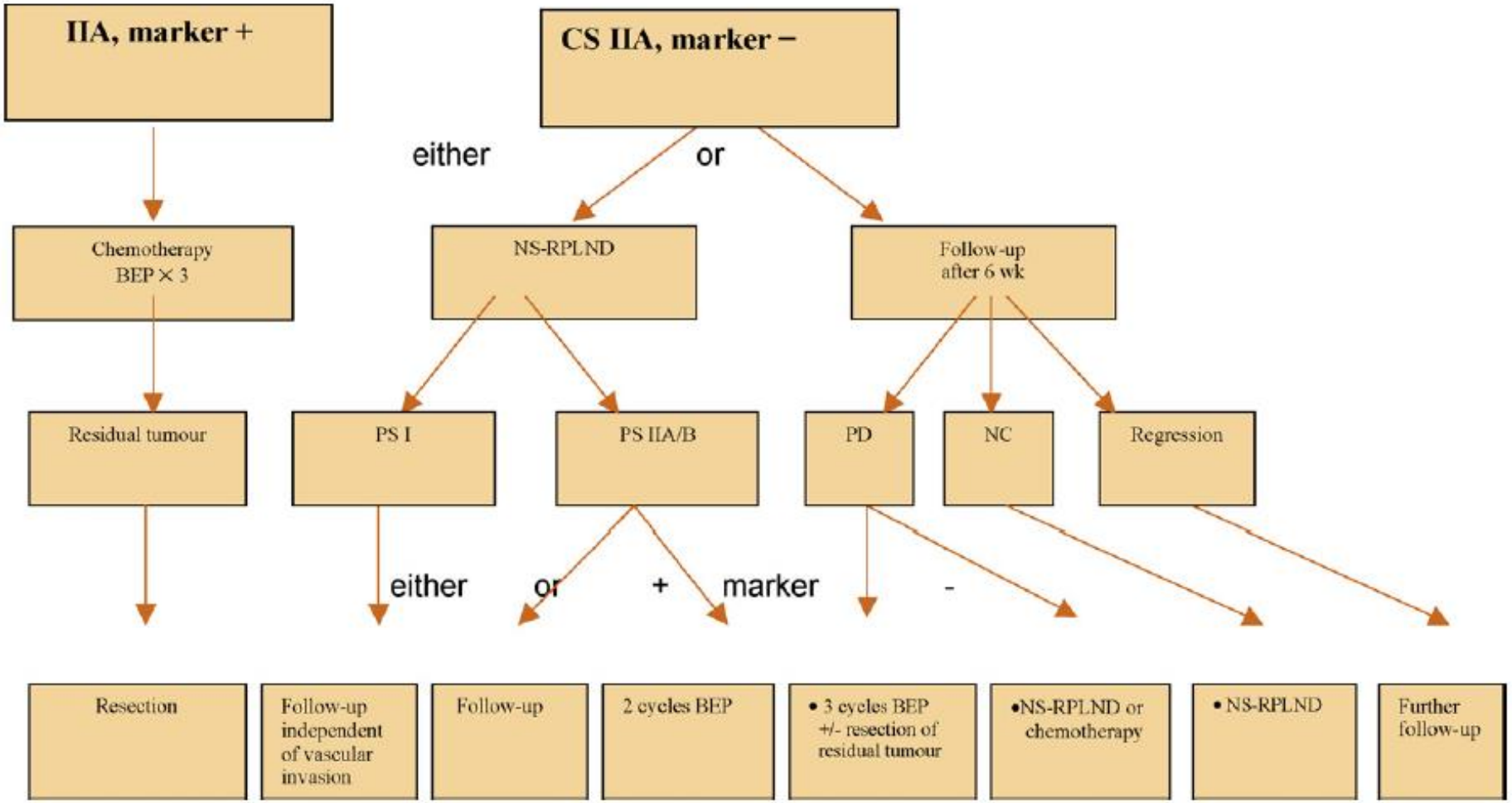
Treatment: Stage IIC, III

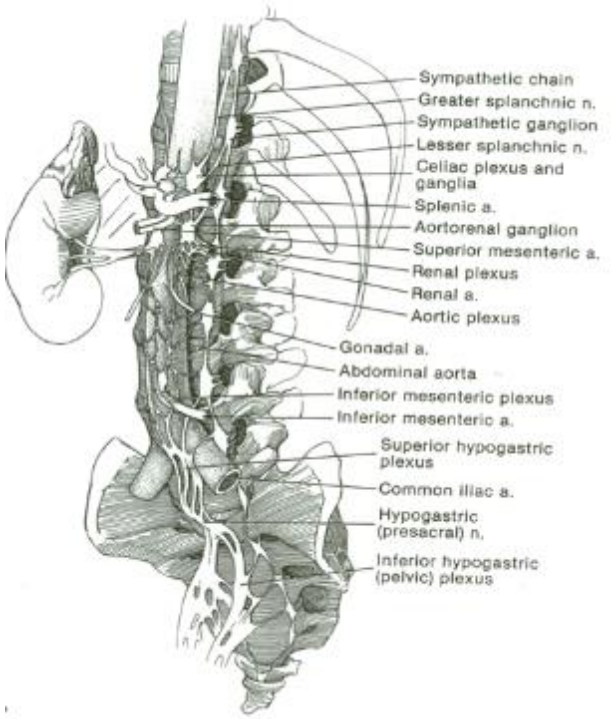
- Cisplatin based chemo (4 cycles of EP or 3 cycles of BEP)
- 90% will have a complete response
- Residual retroperitoneal masses are usually fibrosis
  - RPLND warranted if >3cm and well circumscribed

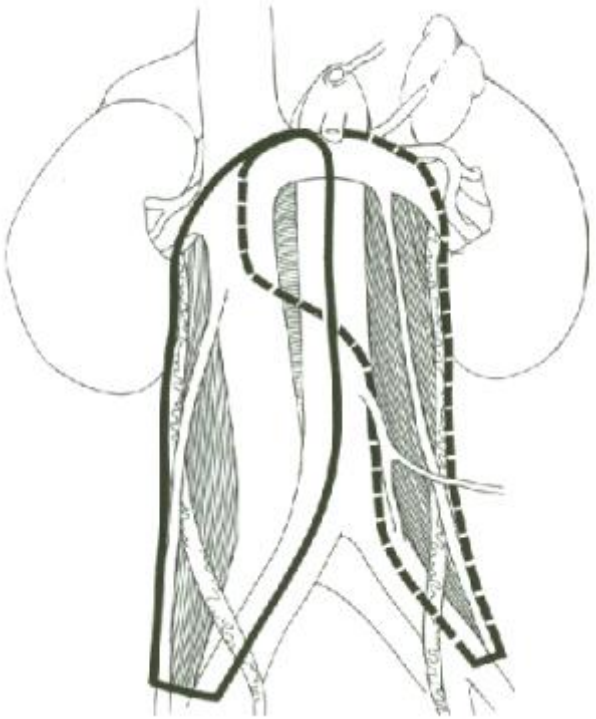
NSGCT


Seminoma/Non-seminoma, advanced disease
Prognostic Group (IGCCCG classification)
Good
Intermediate/ poor
BEP × 3 over 5 d
if conditions against Bleomycin: EP × 4
Schedules:
- Platinum days 1,2 or 1–5
- Etoposide days 1–3 or 1–5
BEP × 4
5-d schedule only
If conditions against bleomycin: PEI (VIP)
repeat cycles d 22 if
no fever/infection
platelets ≥100,000/µl
neutrophils >500/µl
prophylactic G-CSF only if neutropenic fever/infection in previous cycle
Residual tumour
- Marker normalized
- Resectable disease
- Marker elevated but plateau
- Marker increase after short interval (or while CTx)
Follow-up 4–12 wk
No increase of markers
Increase of markers
Resection
Necrosis or teratoma
Complete resection <10% viable tumour cells
>10% viable tumour cells
Incomplete resection of viable tumour
Follow-up
Consolidation chemotherapy (eg, VIP × 2)
Salvage chemotherapy
PEI/VIP; TIP
If platinum not possible, paclitaxel/gemcitabine

NSGCT


IIA, marker +
CS IIA, marker −
either
or
Chemotherapy
BEP × 3
NS-RPLND
Follow-up
after 6 wk
Residual tumour
PS I
PS IIA/B
PD
NC
Regression
either
or
+
marker
-
Resection
Follow-up
independent
of vascular
invasion
Follow-up
2 cycles BEP
• 3 cycles BEP
+/- resection of
residual tumour
•NS-RPLND or
chemotherapy
• NS-RPLND
Further
follow-up





- Major morbidity is ejaculatory dysfunction
- Modified nerve sparing RPLND preserves function in 90-99%
  - Identify the sympathetic nerves
  - Dissection is limited to below the level of the IMA on the ipsilateral side only

Summary

- Testicular tumors are most common cause of painless testicular enlargement – associated with undecended testes and testicular dysgenesis
- 95% are germ cell tumors which may be pure (60%) or mixed (40%).
- The most common pure type is Seminoma
- Clinically the distinction between seminoma and non-seminatous types is important
  - Seminoma – remains confined to testes for long time, spreads mainly to paraaortic nodes
  - Nonseminomatous – tend to spread earlier by both lymphatics and blood.

Questions?

Thank you!